CIP-A107 SOC Laboratory

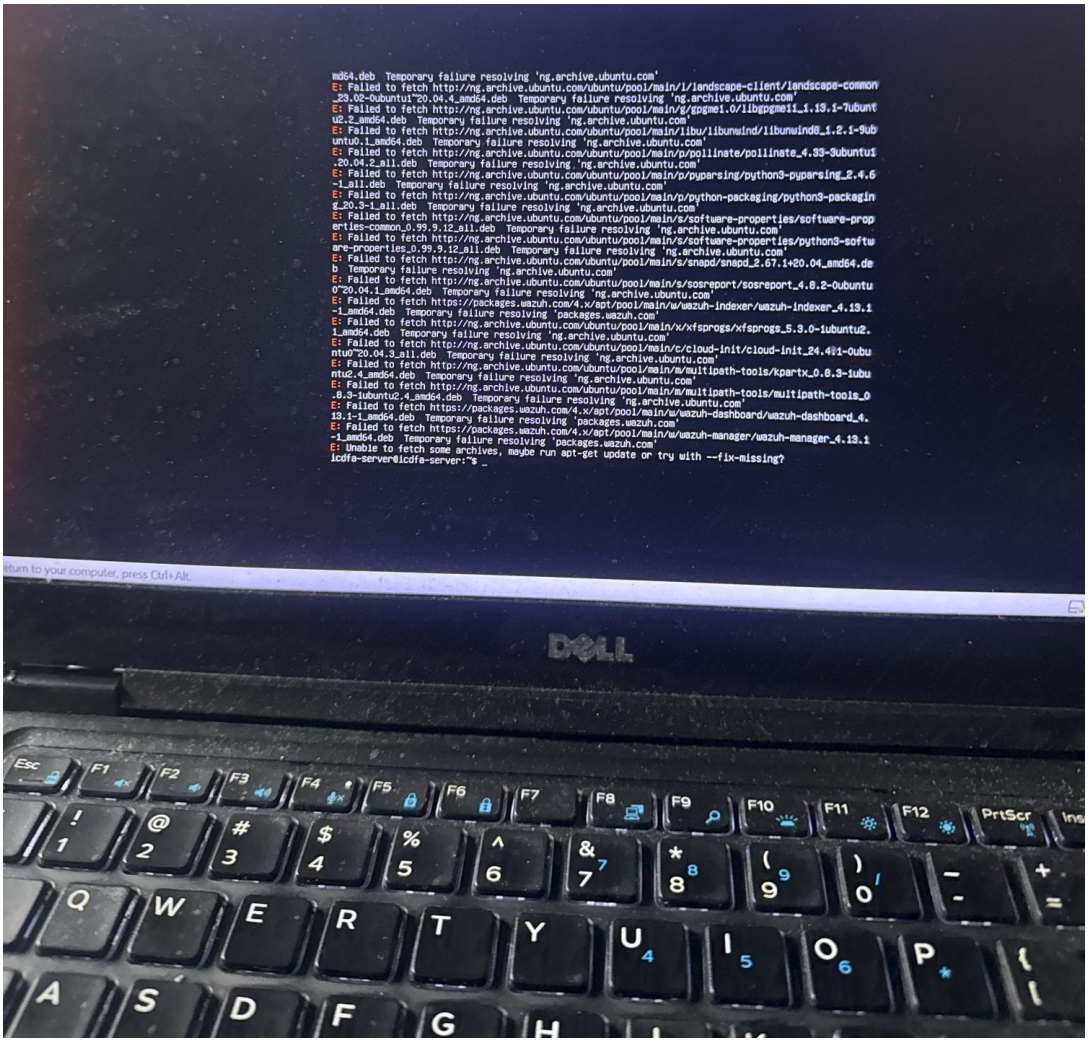
Evidence Appendix

Screenshots and text exports supporting the Technical Report

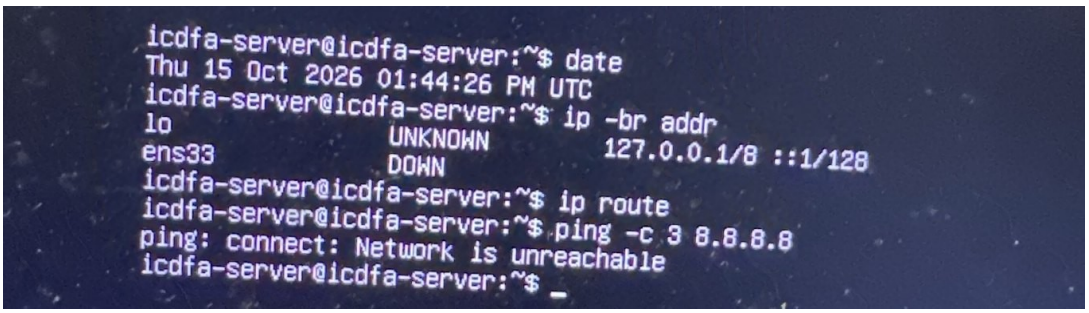
Student	Perry Koko, C1126EHIT17343
Contents	A environment and installation attempts (E01 to E31); B network and access (E32 to E41); C agent registration (E42 to E44); D FIM alerts (E45 to E47); T text exports (T01 to T13).
Capture method	Screenshots are phone photos of the VM console and host screenshots, cropped. The time in each caption is the clock visible in the image; "not visible" means no clock is shown and the date is the day the photo was received. Every file has a SHA-256 in SHA256SUMS.txt and screenshots_manifest.csv.
Redactions	The Wazuh administrator password (E27) and the agent key (E42) are covered. Two Inventory screenshots with host and hardware identifiers were left out.

A. Environment and installation attempts

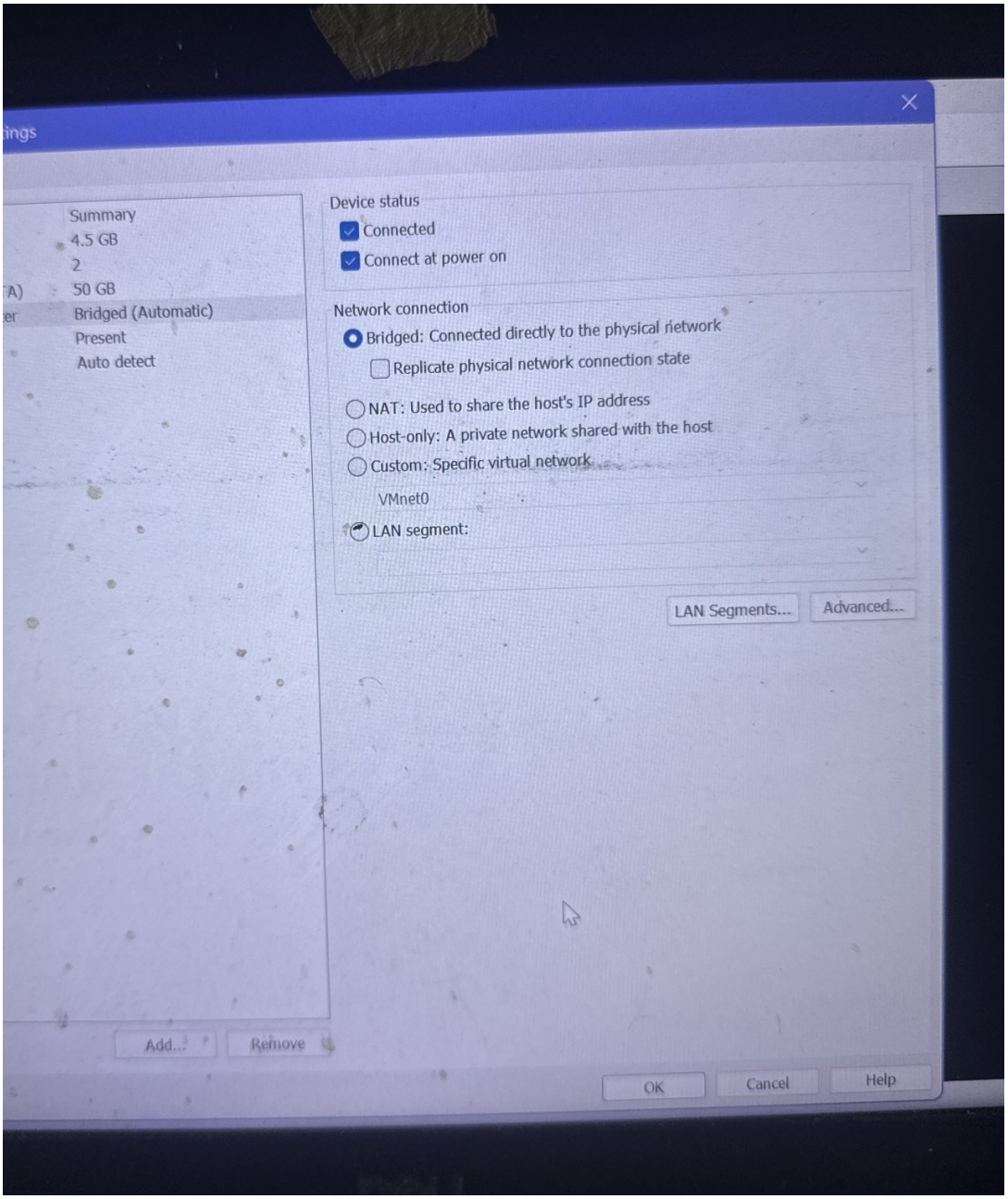
Network faults, five installer runs, resources, snapshot, apt tuning, package pre-download and the service state after run 5.



Ex. E01 *| not visible (15 Oct) | vm-console | apt fails during the first attempt: "Temporary failure resolving" for ng.archive.ubuntu.com and packages.wazuh.com, then "Unable to fetch some archives". | SHA-256 35a8b73dc0b5d8ce…*



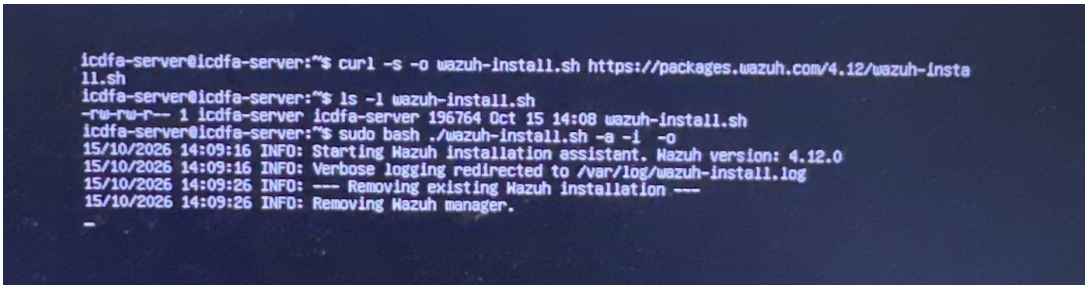
Ex. E02 | *13:44:26 UTC (15 Oct)* | *vm-console* | *`date`, `ip -br addr` (ens33 DOWN) and `ping 8.8.8.8` ("Network is unreachable").* | *SHA-256 d7b2902fe46f9ac8…*



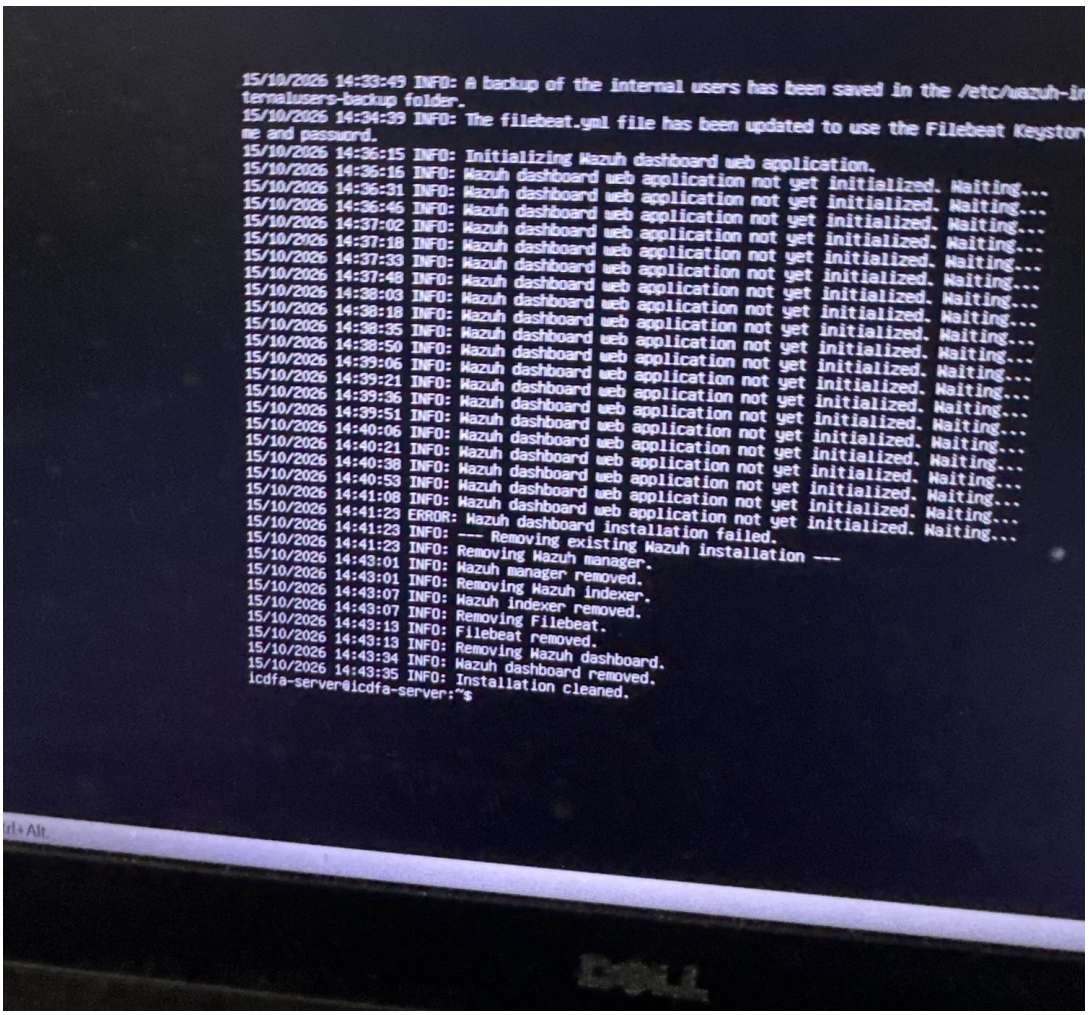
Ex. E03 | *not visible (15 Oct)* | *vmware* | *VM settings before the change: network adapter set to "Bridged (Automatic)".* | *SHA-256 da63af6707a01292…*



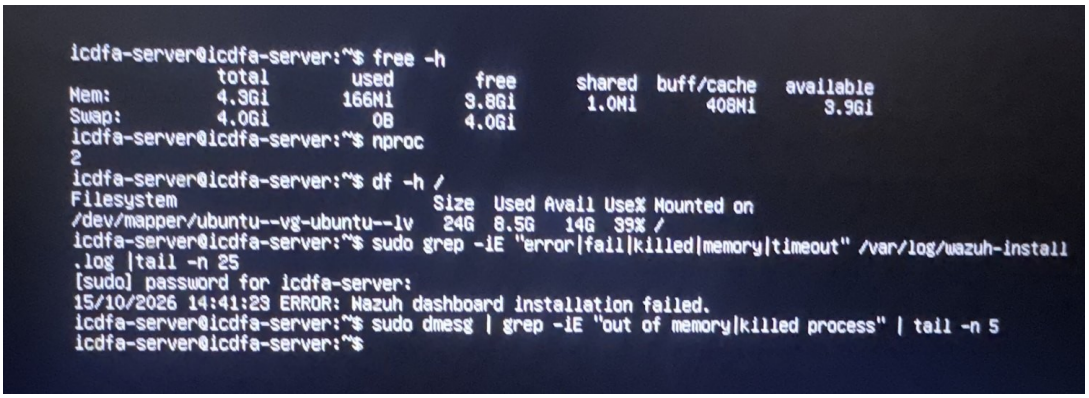
Ex. E04 | *not visible (15 Oct)* | *vm-console* | *`dhclient` on ens33 obtains 192.168.152.131 from DHCP server 192.168.152.254 (NAT); ping to 8.8.8.8 succeeds, 3 of 3.* | *SHA-256 e591accb3f8c50de…*



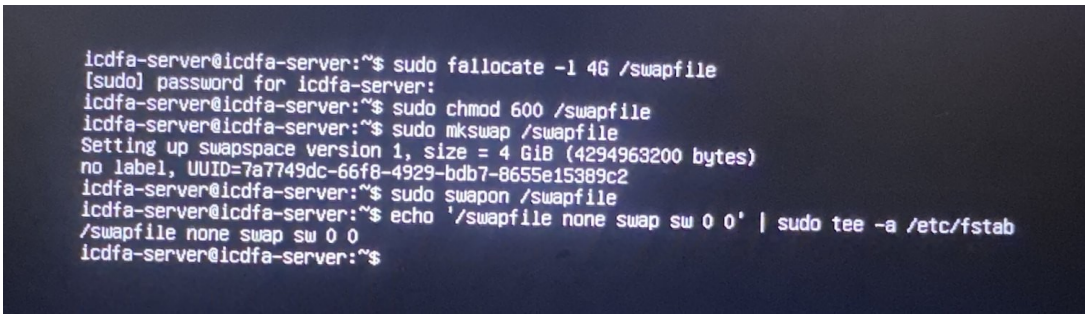
Ex. E05 | *14:09:16 UTC (15 Oct)* | *vm-console* | *Installer downloaded (`curl`) and run 1 started: "Starting Wazuh installation assistant. Wazuh version: 4.12.0".* | *SHA-256 1fc62304f51137b1…*



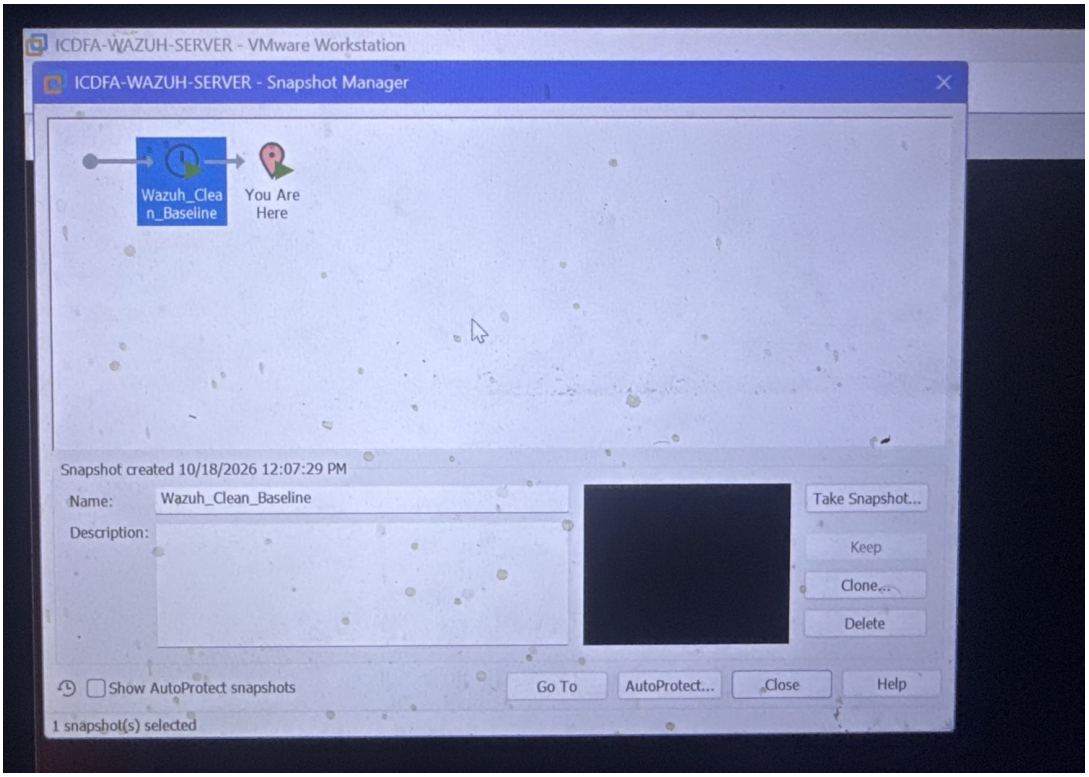
Ex. E06 | *14:36:15 to 14:43:35 UTC (15 Oct)* | *vm-console* | *Run 1 fails: "Wazuh dashboard web application not yet initialized. Waiting…" until the ERROR at 14:41:23, then rollback ("Installation cleaned").* | *SHA-256 a5648b49c1f401b2…*



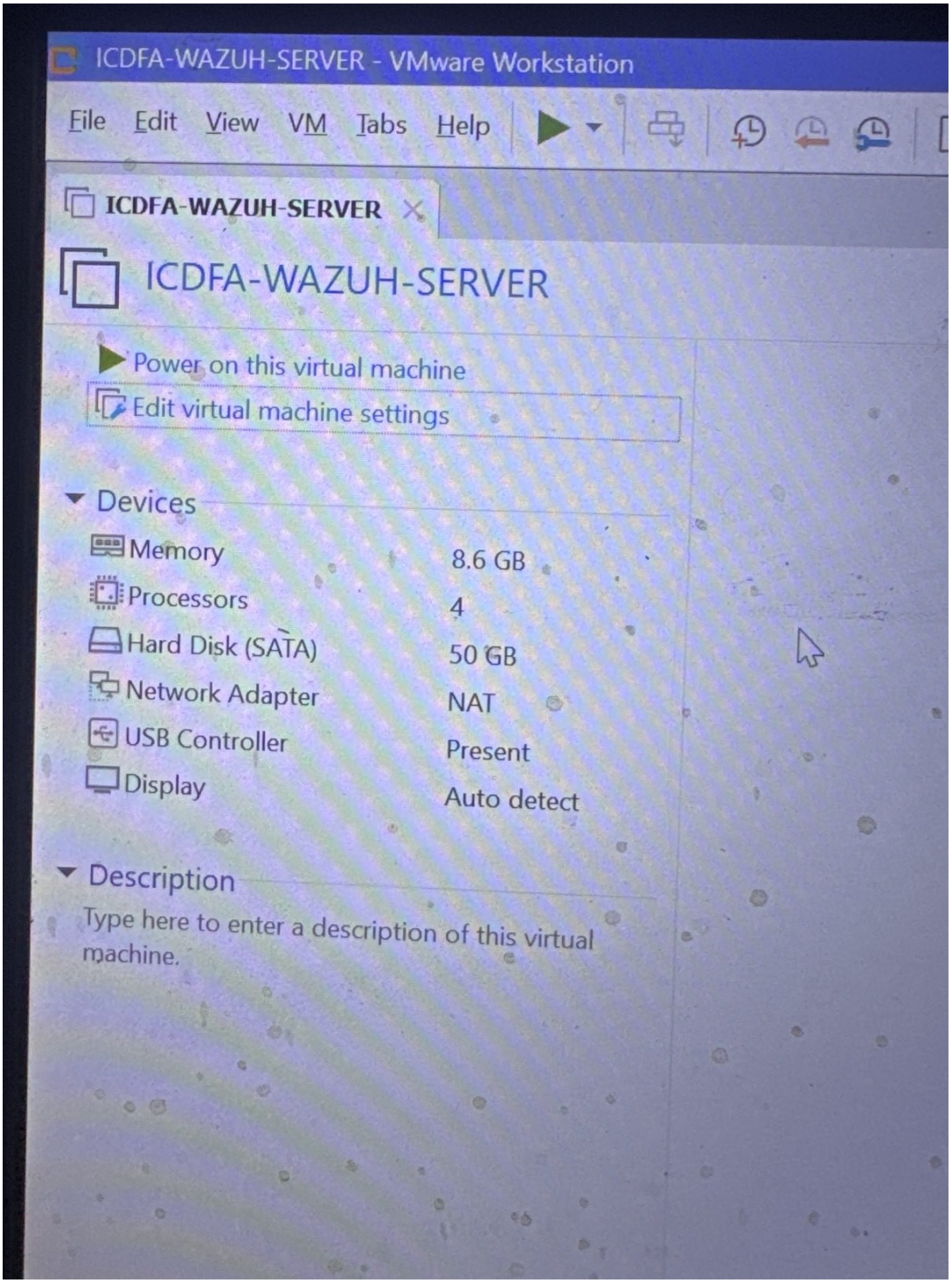
Ex. E07 | *after 14:43 UTC (15 Oct)* | *vm-console* | *Resource check after run 1: `free -h` 4.3 GiB total, `nproc` 2, `df -h /` 24 GB root; the log grep shows only the dashboard ERROR; the OOM search is empty.* | *SHA-256 495a7cb794e18579…*



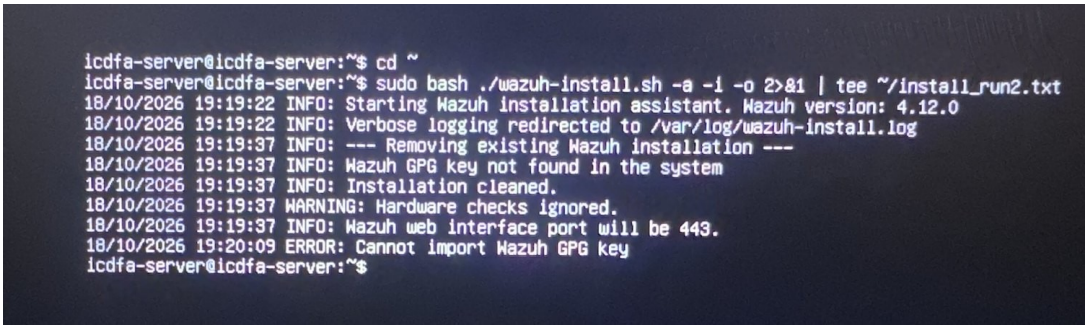
Ex. E08 | *not visible (18 Oct)* | *vm-console* | *Extra swap file created: `fallocate -l 4G /swapfile`, `chmod 600`, `mkswap`, `swapon`, entry added to /etc/fstab.* | *SHA-256 3d45a774470ca9c4…*



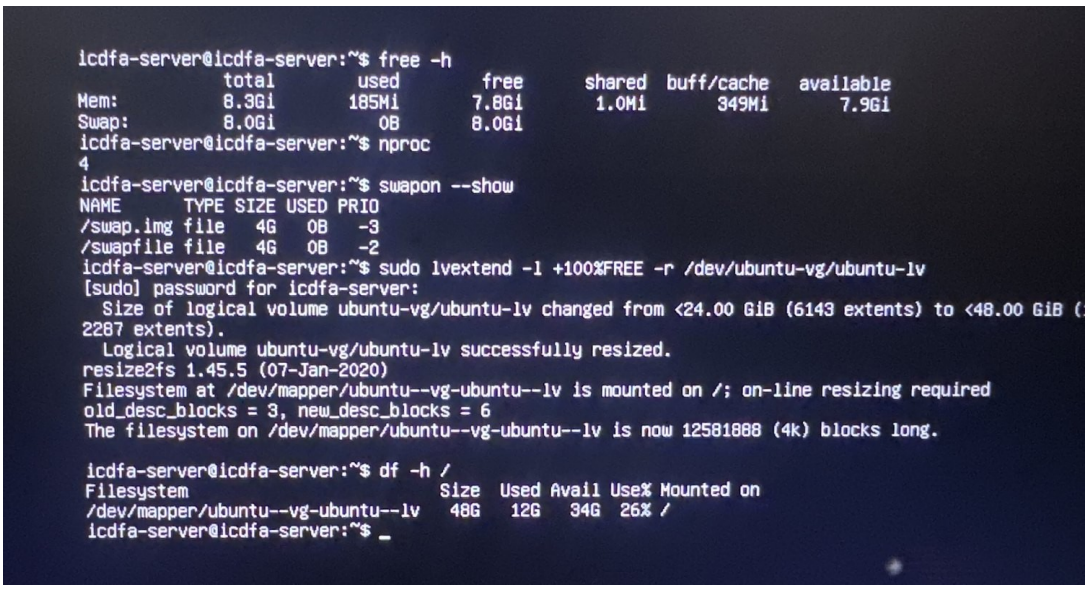
Ex. E09 | *12:07:29 PM host clock (19:07:29 UTC, 18 Oct) | vmware | Snapshot Manager: snapshot "Wazuh_Clean_Baseline" created before further installation attempts. | SHA-256 1def9fb1f56d31a5…*



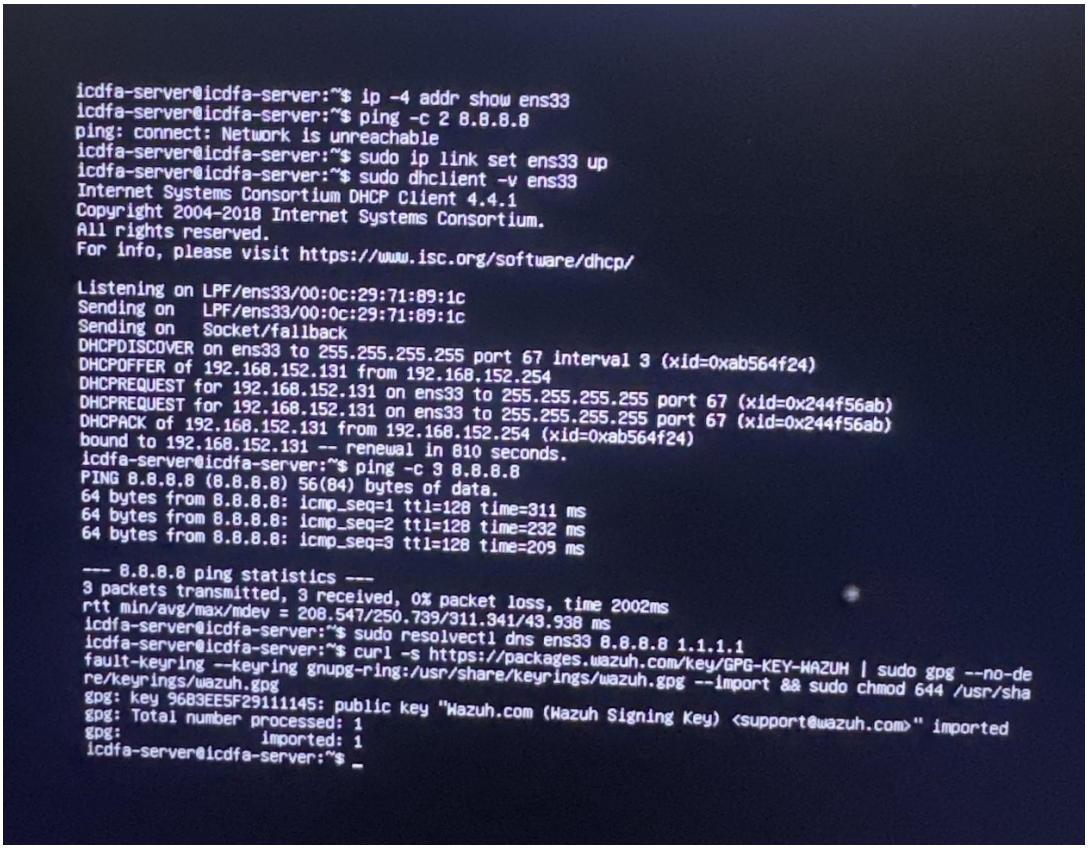
Ex. E10 | *not visible (18 Oct)* | *vmware* | *VM settings after resizing: memory 8.6 GB, 4 processors, hard disk 50 GB, network adapter NAT.* | *SHA-256 162147974c387d18…*



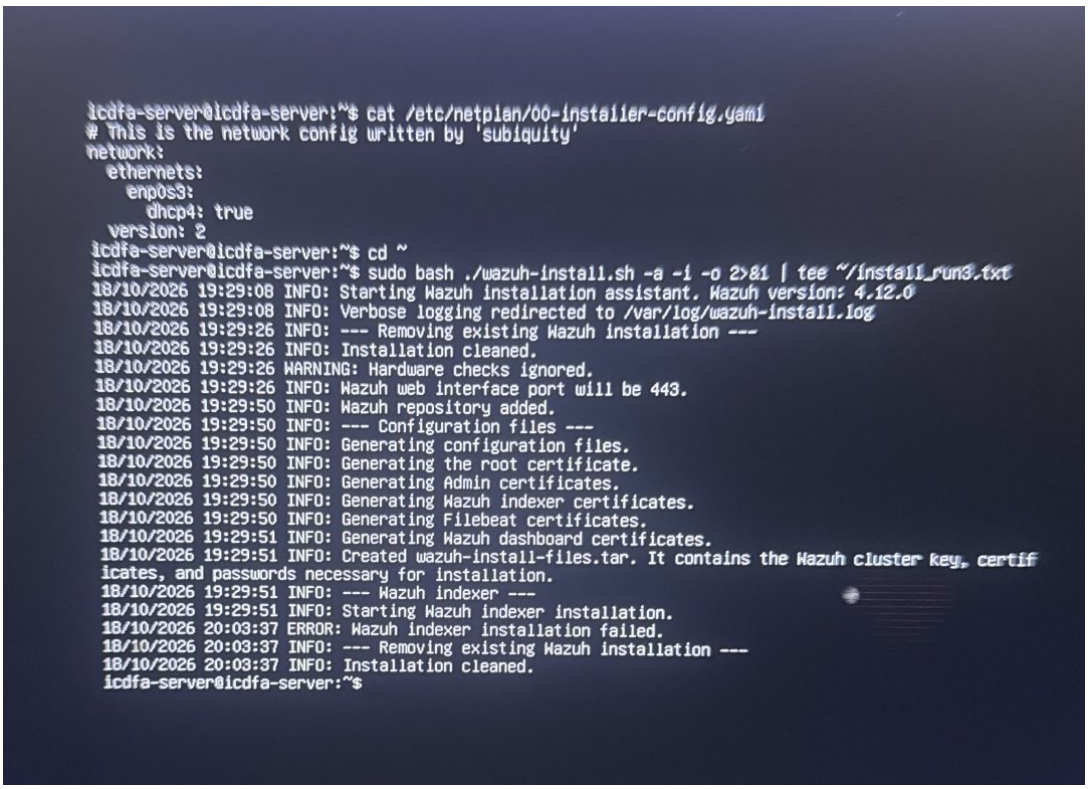
Ex. E11 | *19:19:22 to 19:20:09 UTC (18 Oct)* | *vm-console* | *Run 2: the installer stops with "ERROR: Cannot import Wazuh GPG key".* | *SHA-256 e8191b0bd72da9d4…*



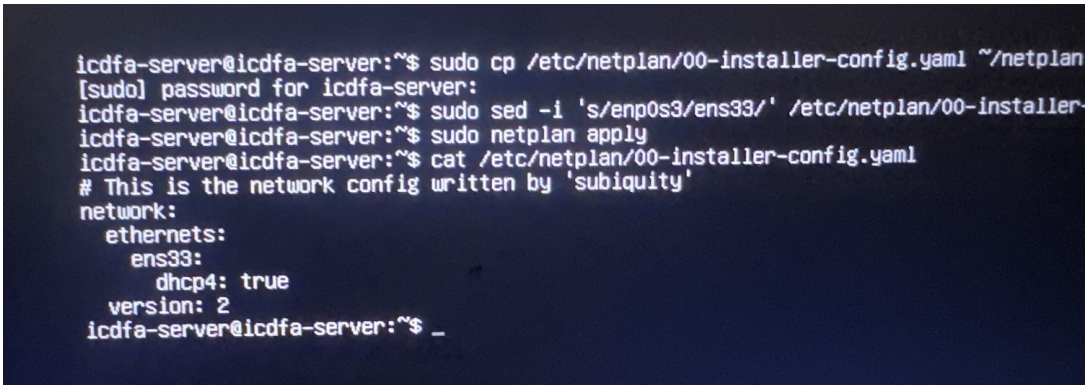
Ex. E12 | *not visible (18 Oct) | vm-console | Resources after the change:* `free -h` *8.3 GiB RAM and 8.0 GiB swap,* `nproc` *4;* `lvextend` *grows the root volume from 24 to 48 GiB,* `resize2fs`*,* `df -h /` *48 GB with 34 GB free. | SHA-256 f3b36446be3c3c44…*



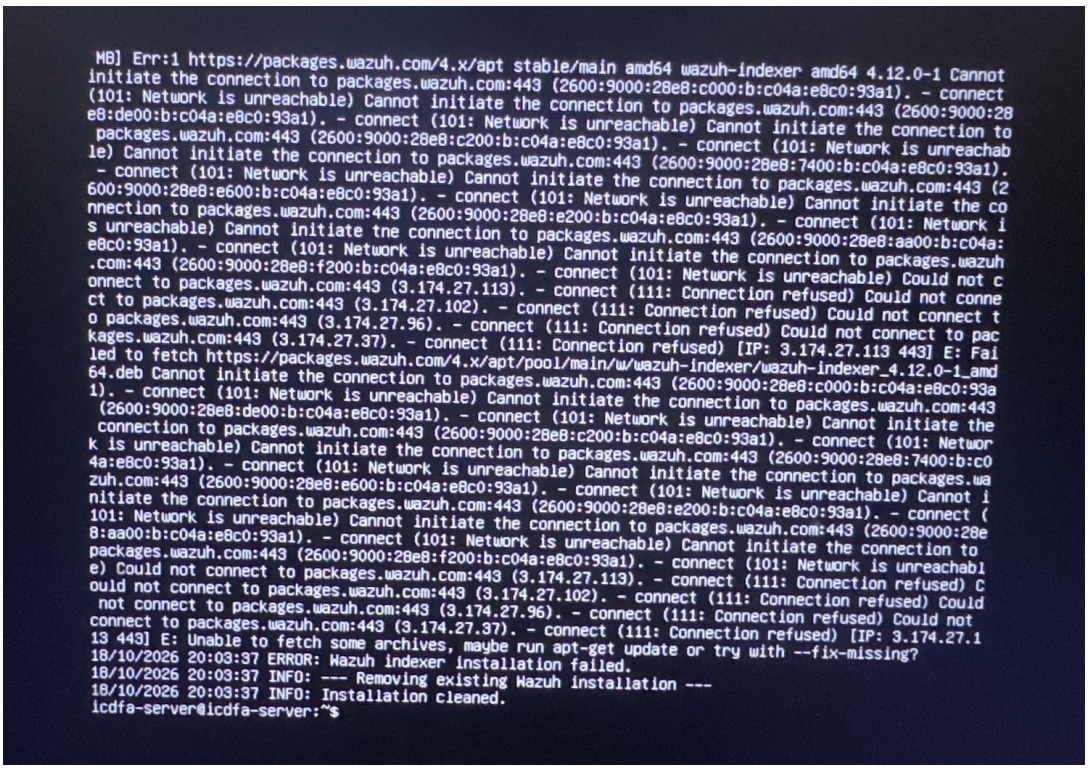
Ex. E13 | *not visible (18 Oct) | vm-console | ens33 had lost its IPv4 address (ping "Network is unreachable");* `dhclient` *leases 192.168.152.131; the Wazuh GPG key is then imported. | SHA-256 4f9946ca9b6ecb7b…*



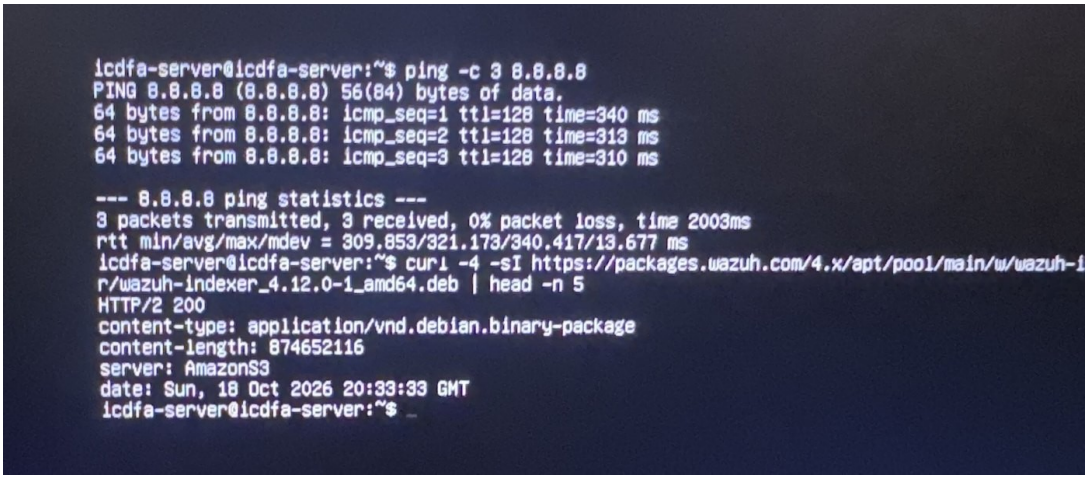
Ex. E14 | *19:29:08 to 20:03:37 UTC (18 Oct) | vm-console | Netplan file still names the interface `enp0s3`; run 3 starts at 19:29:08 and fails at 20:03:37 with "ERROR: Wazuh indexer installation failed", followed by rollback. | SHA-256 0b1ce22d340d9a91…*



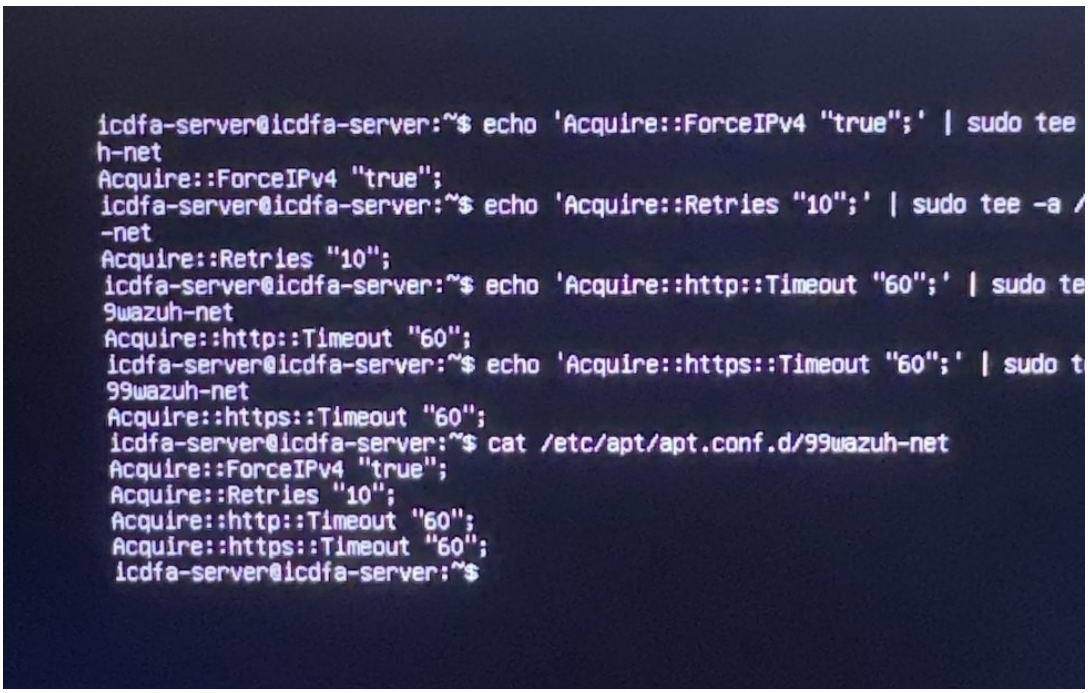
Ex. E15 | *not visible (18 Oct) | vm-console | Netplan repair: `sed -i s/enp0s3/ens33/`, `netplan apply`; the file now defines `ens33` with `dhcp4: true`. | SHA-256 ce1043afccaab49e…*



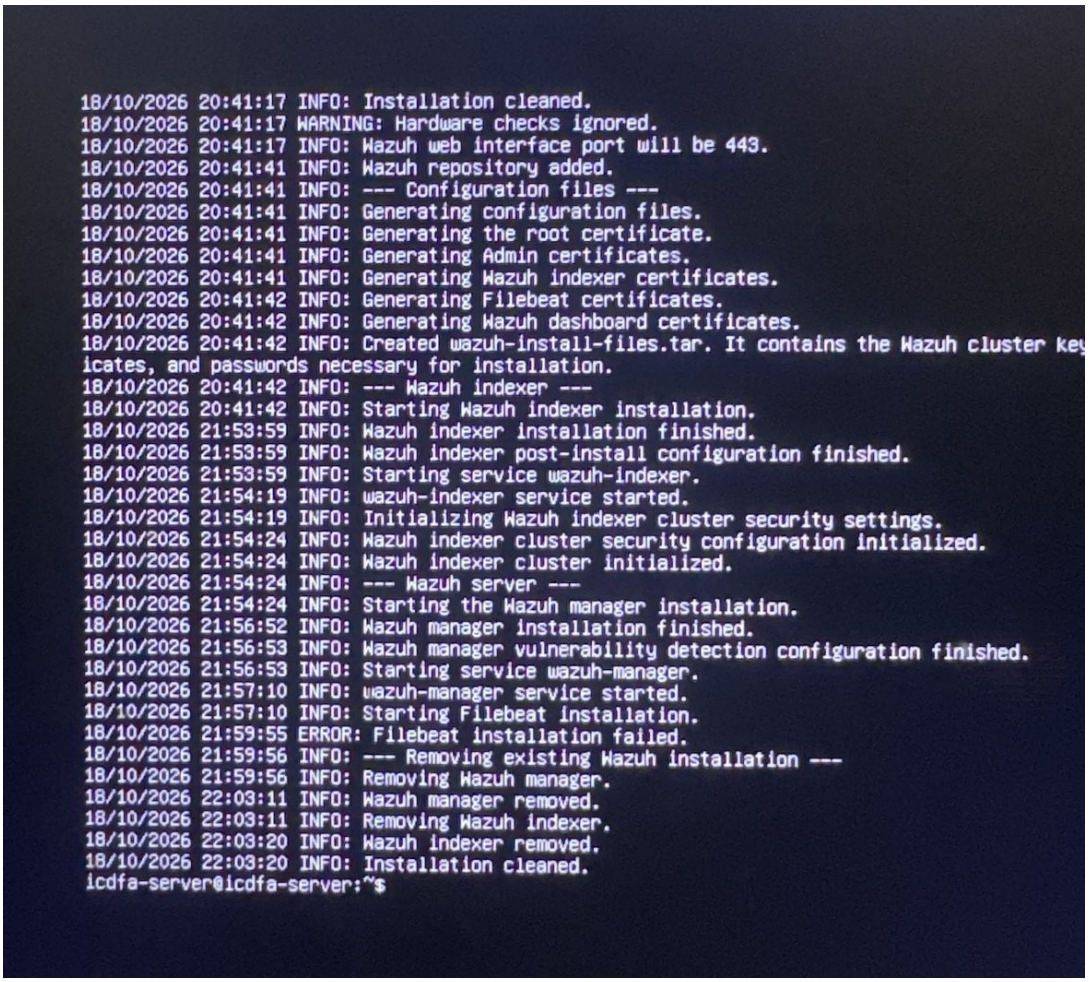
Ex. E16 | *around 20:03 UTC (18 Oct) | vm-console | apt errors for the indexer package: "Cannot initiate the connection to packages.wazuh.com:443 (2600:9000:…)", "Network is unreachable" and "Connection refused" for 3.174.27.x. | SHA-256 1bb81ed9523624fd…*



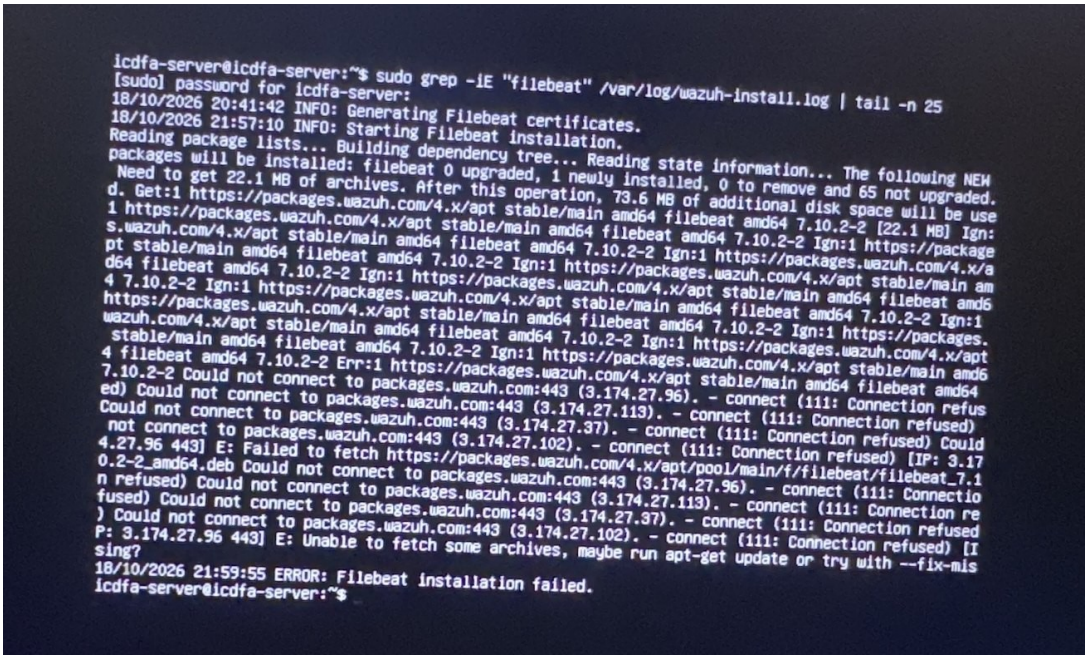
Ex. E17 | *20:33:33 GMT (18 Oct) | vm-console | Connectivity re-check: ping 8.8.8.8 succeeds; `curl -sI` of the indexer package returns HTTP/2 200, content-length 874,652,116 bytes. | SHA-256 afc06df8f71228eb…*



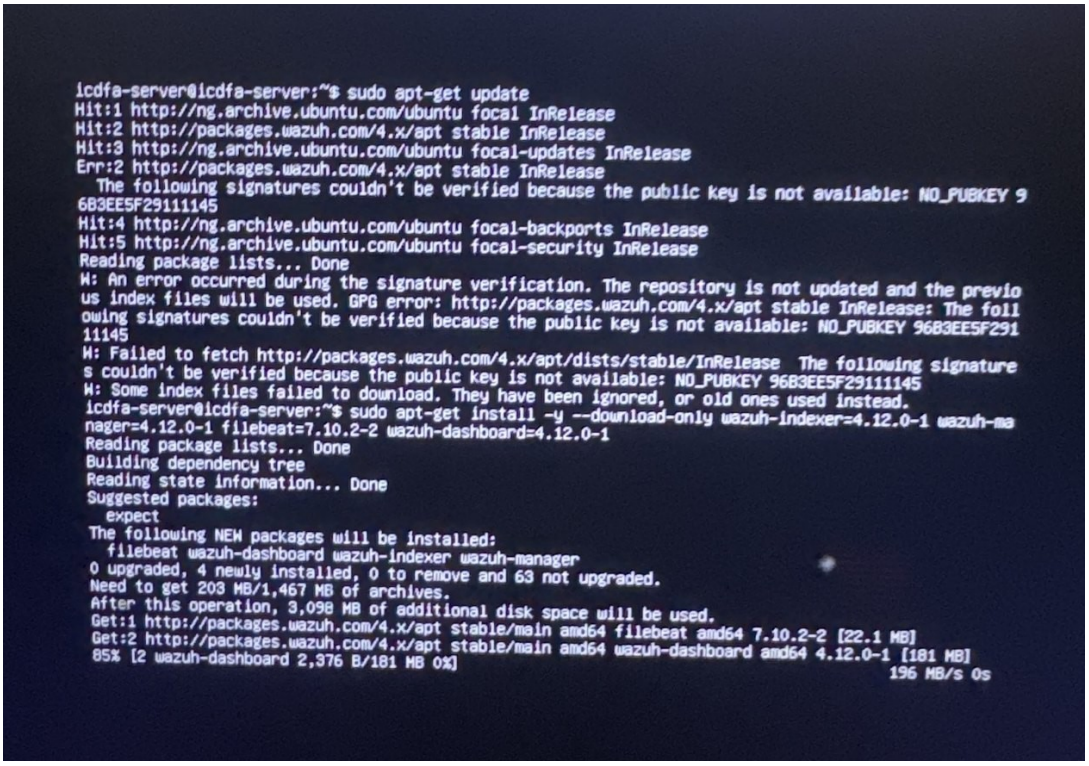
Ex. E18 | *not visible (18 Oct) | vm-console | apt resilience file /etc/apt/apt.conf.d/99wazuh-net: ForceIPv4 true, Retries 10, http and https Timeout 60. | SHA-256 b7468f2649717900…*



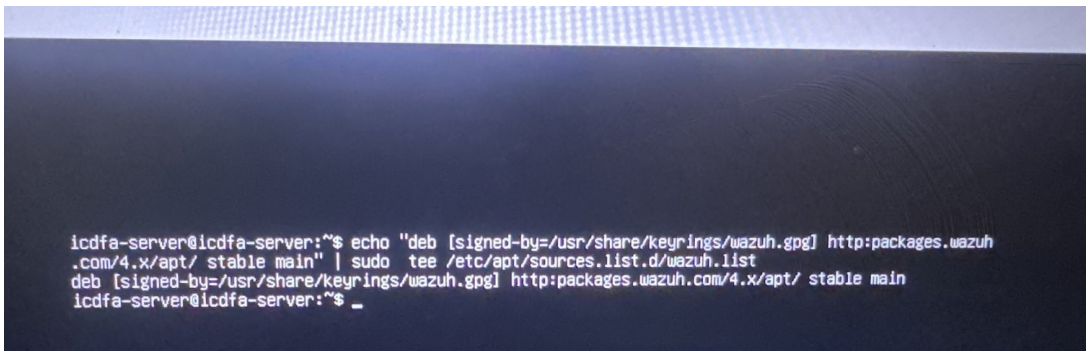
Ex. E19 | *20:41:17 to 22:03:20 UTC (18 Oct) | vm-console | Run 4: indexer installation started 20:41:42 and finished 21:53:59; manager finished 21:56:52; "ERROR: Filebeat installation failed" at 21:59:55; rollback to 22:03:20. | SHA-256 c0941c2d00101c89…*



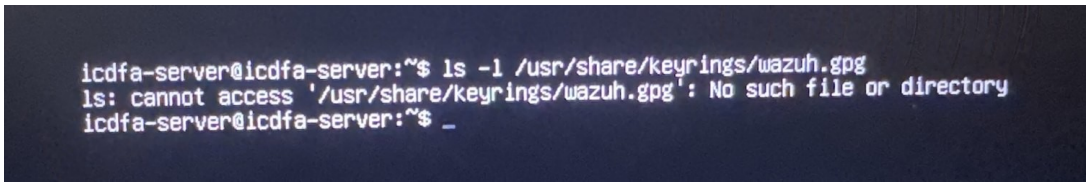
Ex. E20 | *21:59:55 UTC (18 Oct)* | *vm-console* | *Install-log search for "filebeat": "Connection refused" to packages.wazuh.com, then "Filebeat installation failed".* | *SHA-256 80280062df43819d…*



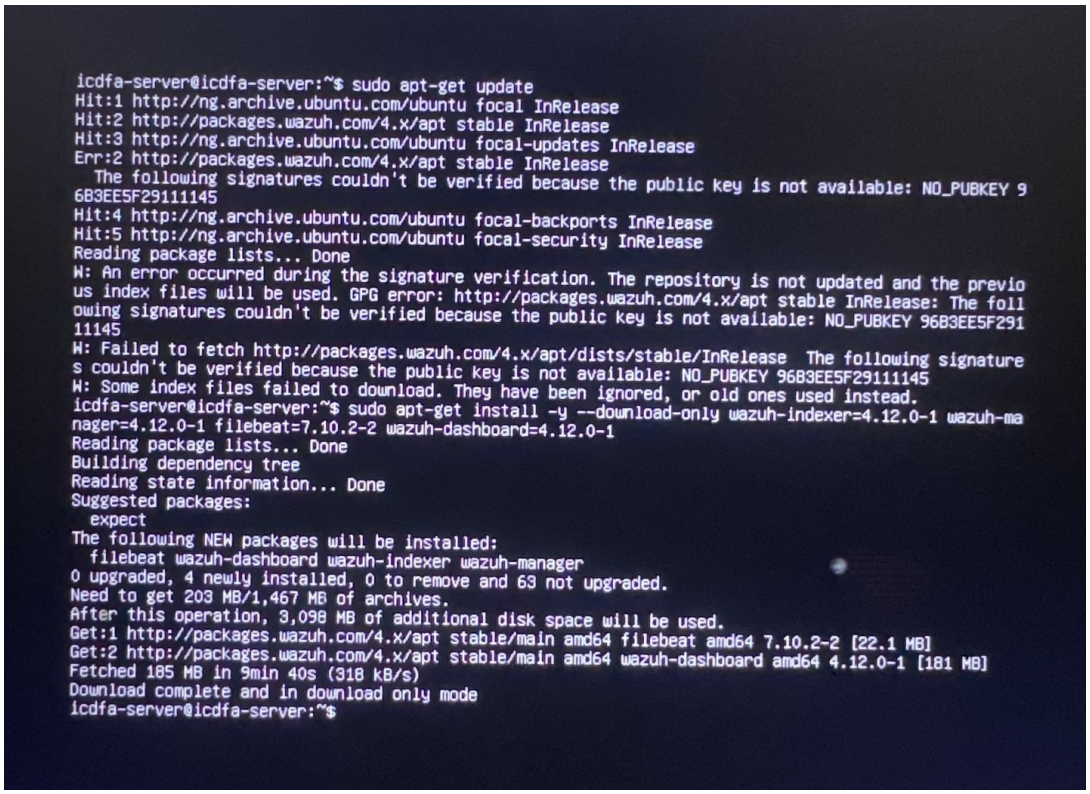
Ex. E21 | *not visible (18 Oct)* | *vm-console* | *`apt-get update` warns NO_PUBKEY 9683EE5F29111145 after the rollback removed the key; `apt-get install --download-only` starts (203 MB to fetch).* | *SHA-256 6f4c570eaa0e6458…*



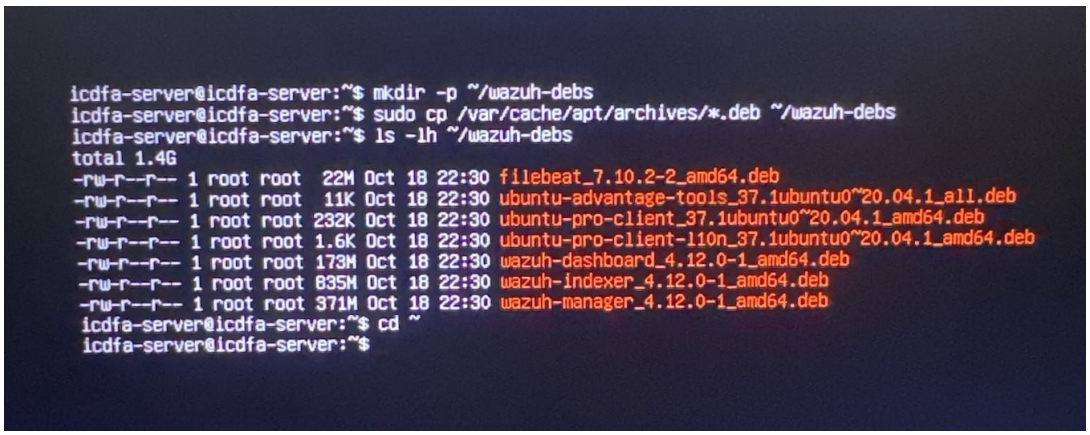
Ex. E22 | *not visible (18 Oct)* | *vm-console* | *Manual repository line written to /etc/apt/sources.list.d/wazuh.list (the URL was typed without "//", so the installer later rewrote the repository).* | *SHA-256 8741a853c03b9b62…*



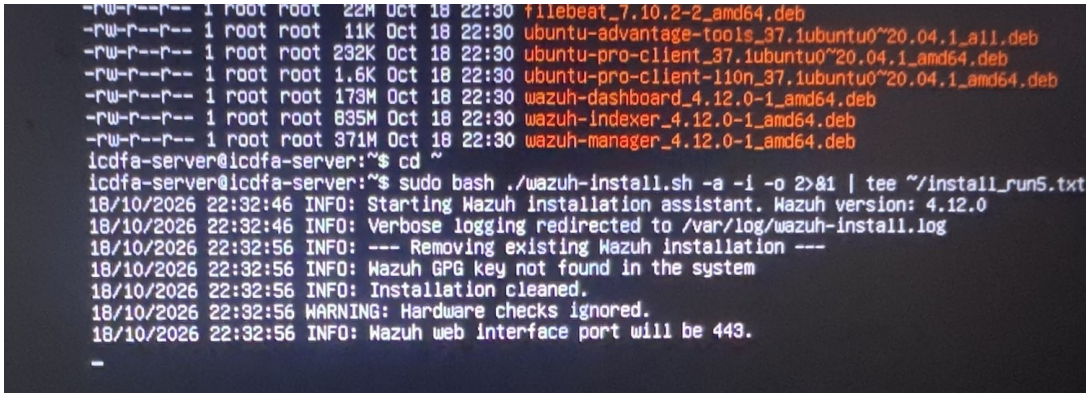
Ex. E23 | *not visible (18 Oct)* | *vm-console* | *`ls -l /usr/share/keyrings/wazuh.gpg`: "No such file or directory" (key removed by the rollback; the installer re-adds it).* | *SHA-256 653e88a8d6e70057…*



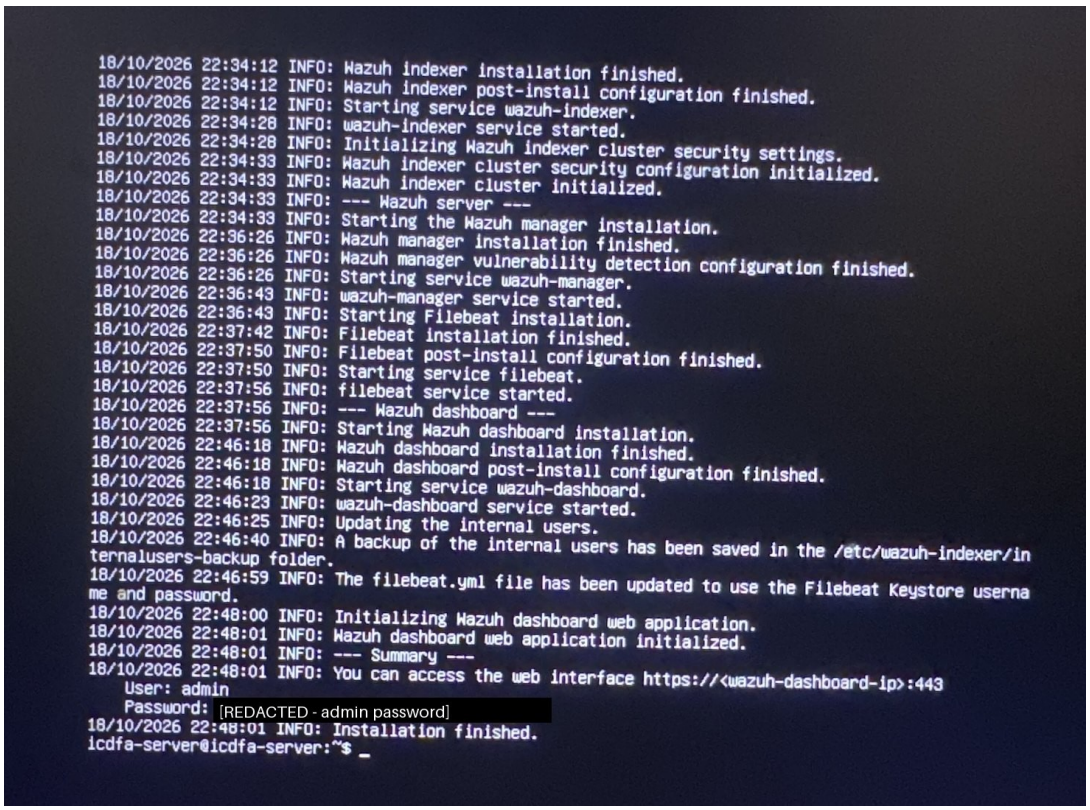
Ex. E24 | *about 22:30 UTC (18 Oct)* | *vm-console* | *Download-only finished: "Fetched 185 MB in 9min 40s (318 kB/s)".* | *SHA-256 9466f98808461a83…*



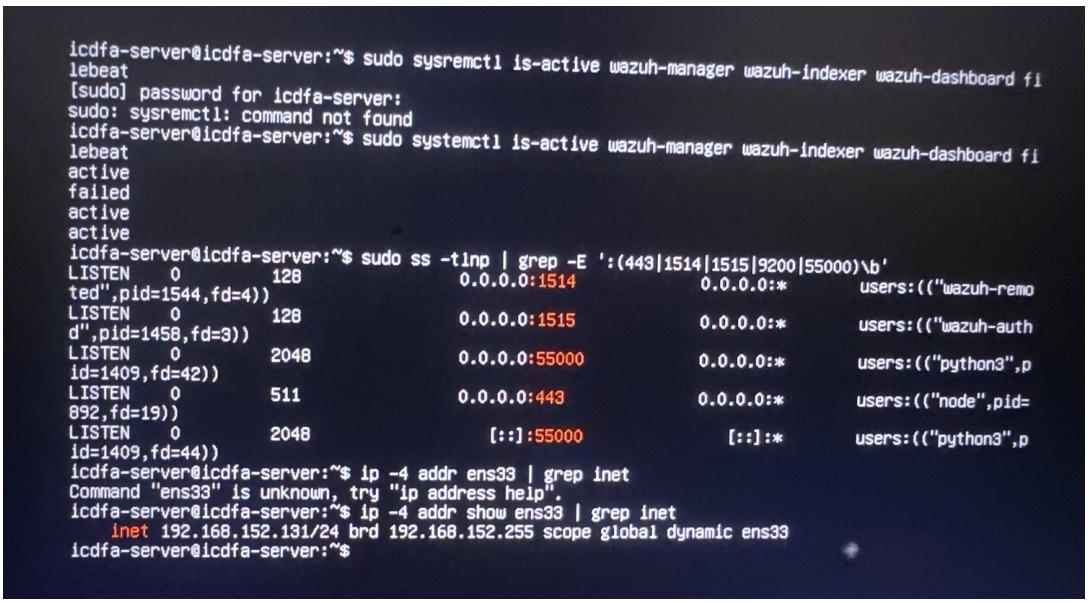
Ex. E25 | *22:30 UTC (18 Oct) | vm-console | Packages copied to ~/wazuh-debs (`mkdir`, `cp`, `ls -lh`): 1.4 GB including wazuh-indexer, wazuh-manager, wazuh-dashboard and filebeat packages. | SHA-256 7976a545ffda20dd…*



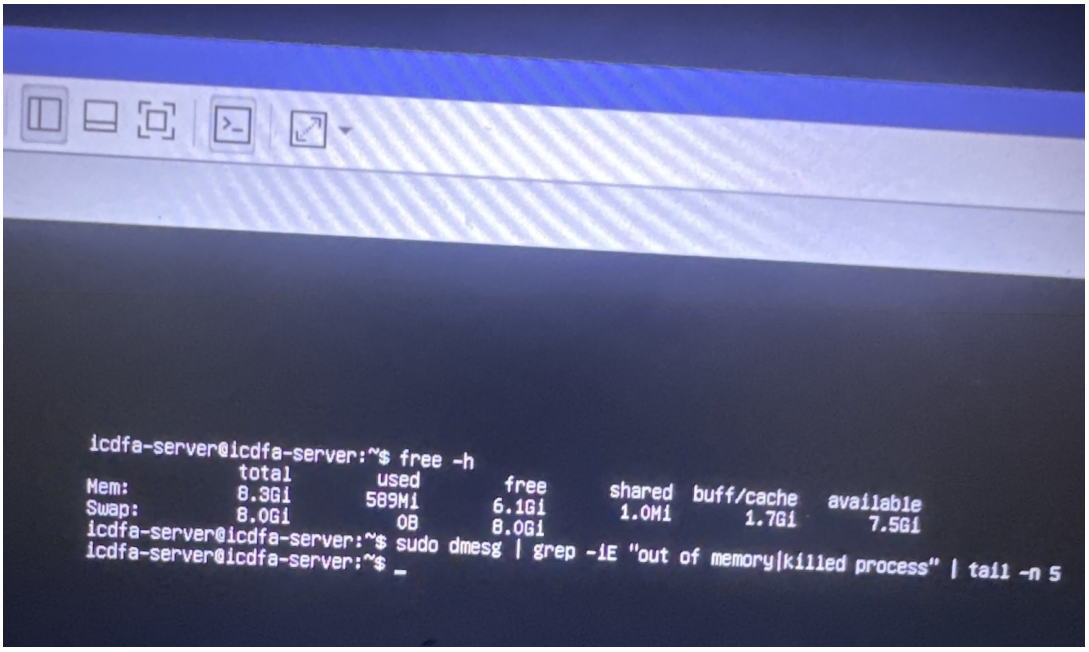
Ex. E26 | *22:32:46 UTC (18 Oct) | vm-console | Run 5 starts: `sudo bash ./wazuh-install.sh -a -i -o 2>&1 | tee ~/install_run5.txt`; "Hardware checks ignored"; "Wazuh web interface port will be 443". | SHA-256 d4af7d0c231886b0…*



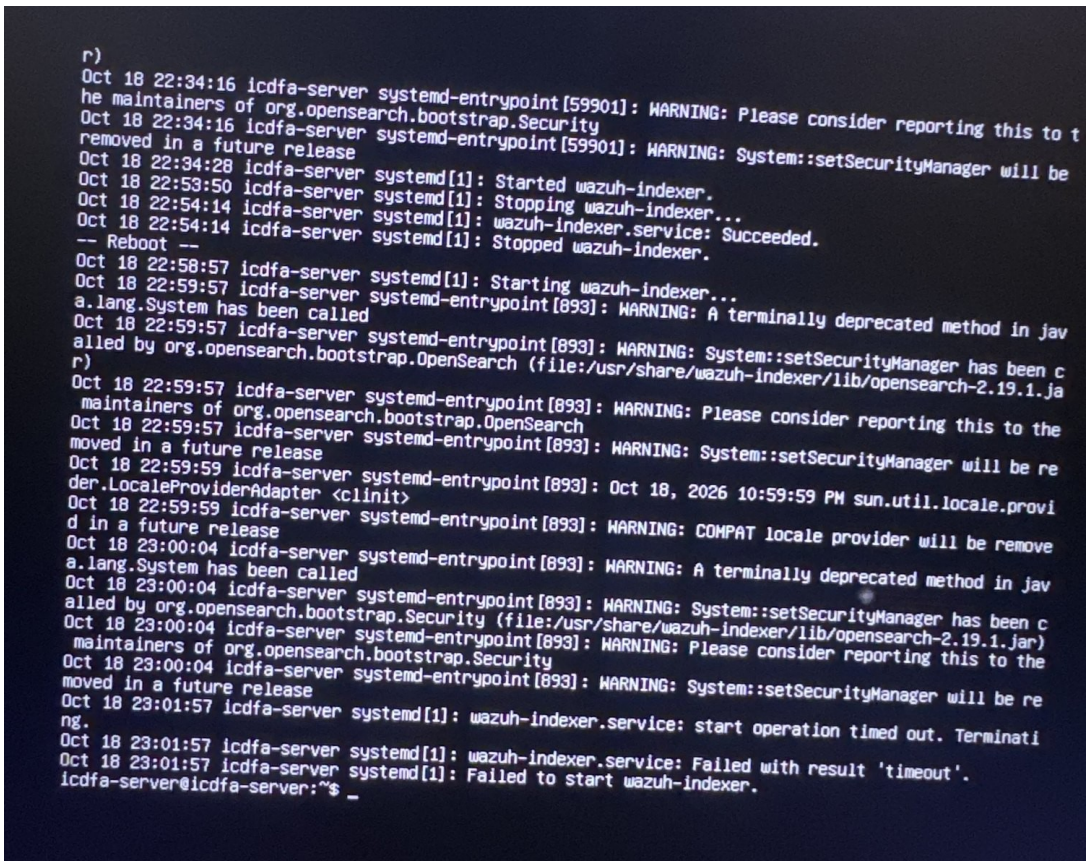
Ex. E27 | *22:34:12 to 22:48:01 UTC (18 Oct) | vm-console | Run 5 completes: indexer, manager, Filebeat and dashboard installed; "Installation finished" at 22:48:01. The administrator password line is covered on purpose. | SHA-256 79d0233dedf1ebeb…*



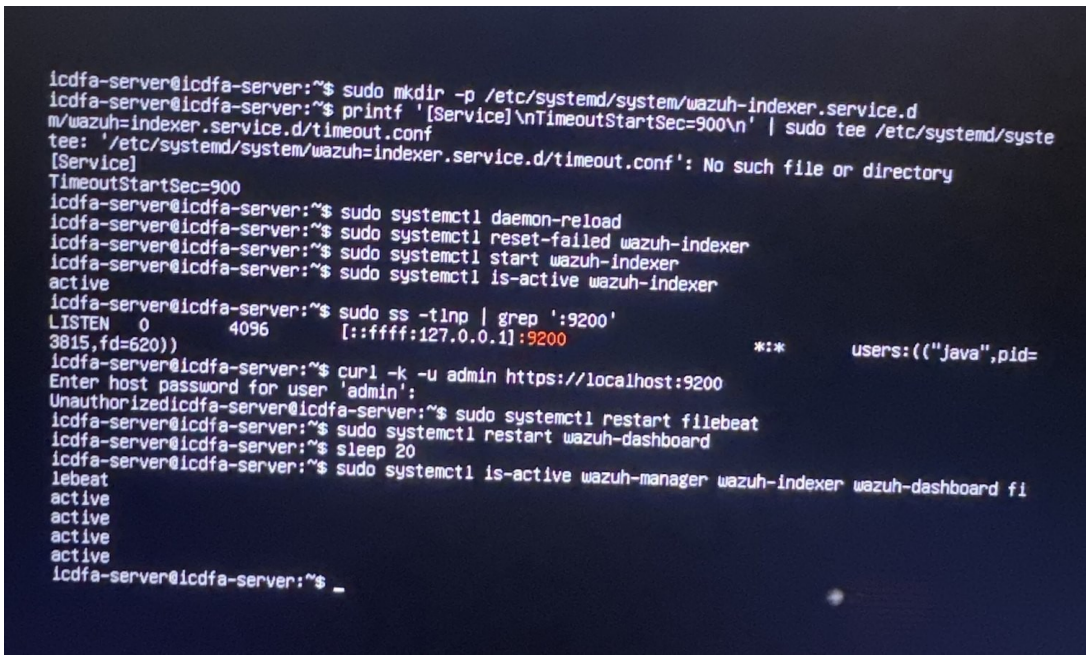
Ex. E28 | *after 22:48 UTC (18 Oct)* | *vm-console* | *Service check: wazuh-manager active, wazuh-indexer failed, wazuh-dashboard active, filebeat active; listening ports 1514, 1515, 55000 and 443; ens33 192.168.152.131.* | *SHA-256 b85f817afda49aa1…*



Ex. E29 | *around 23:00 UTC (18 Oct)* | *vm-console* | *Memory check: 8.0 GiB total, 6.1 GiB free; the out-of-memory search in dmesg is empty.* | *SHA-256 ba266c3b1985dfd0…*



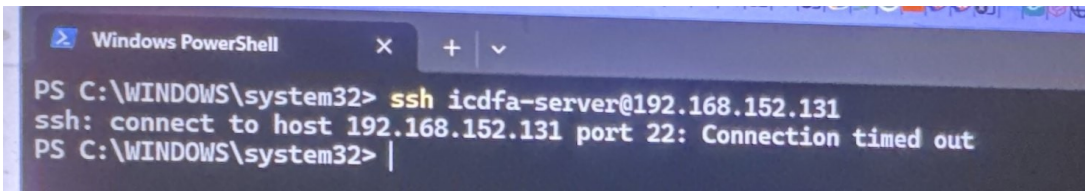
Ex. E30 | *23:01:57 UTC (18 Oct) | vm-console | Journal of wazuh-indexer: "start operation timed out. Terminating", "Failed with result timeout", "Failed to start wazuh-indexer". | SHA-256 5efc663d8ea8ca36…*



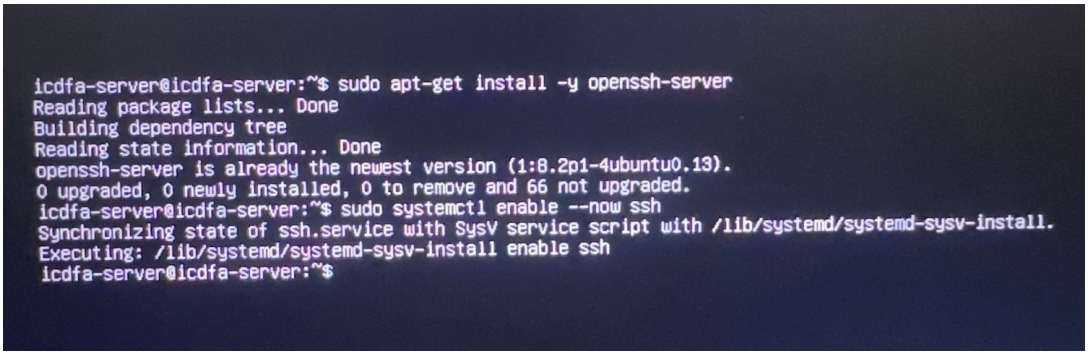
Ex. E31 | *after 23:01 UTC (18 Oct) | vm-console | Indexer started again (`systemctl start`, then `is-active` = active); port 9200 listens on 127.0.0.1 (java); `curl -k -u admin` returns "Unauthorized" because the password was mistyped; all four services active. | SHA-256 8321e88ceea6b132…*

B. Network repair, access and dashboard

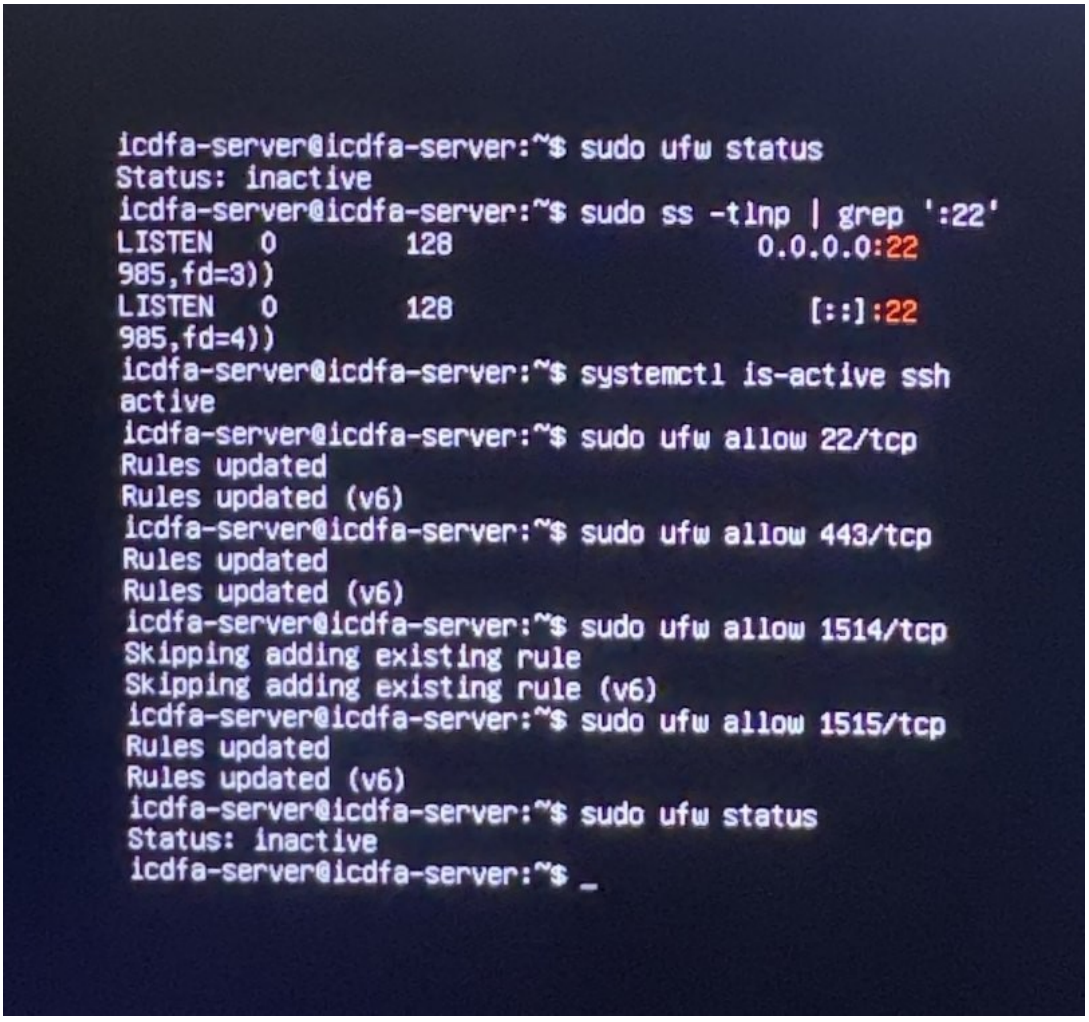
SSH and VMnet8 problem, the new address, connectivity tests, login page and overview.



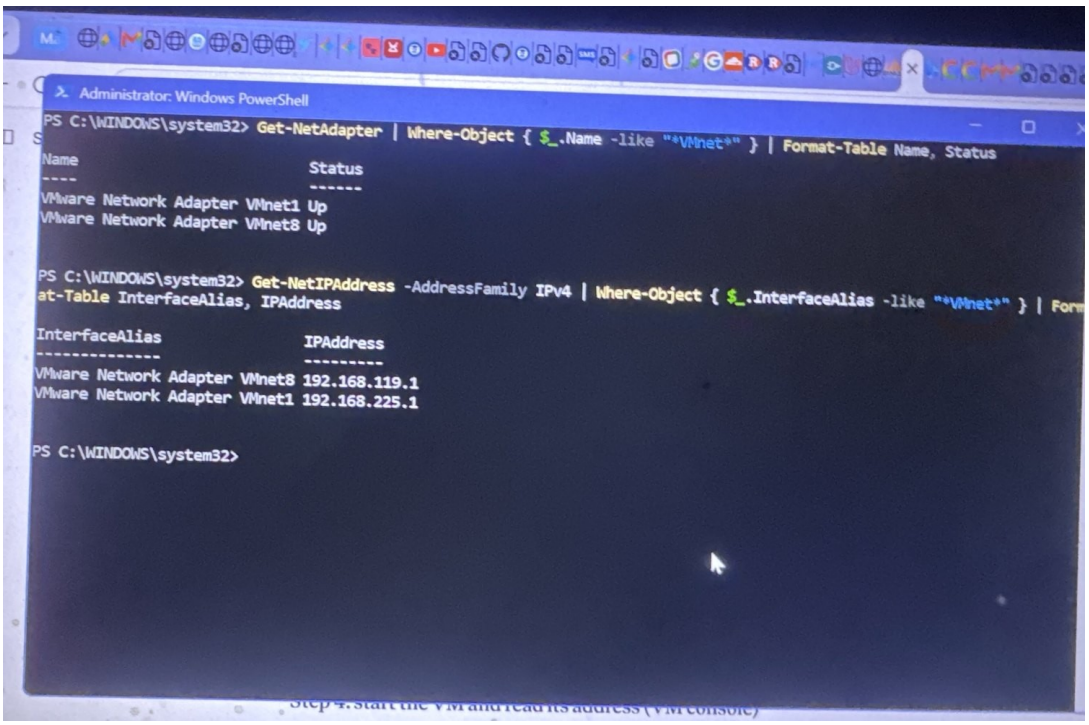
Ex. E32 | *not visible (18 Oct)* | *powershell* | *Windows host:* `ssh icdfa-server@192.168.152.131` *ends with "connect to host … port 22: Connection timed out".* | *SHA-256 cd33d2cd81f88ce2…*



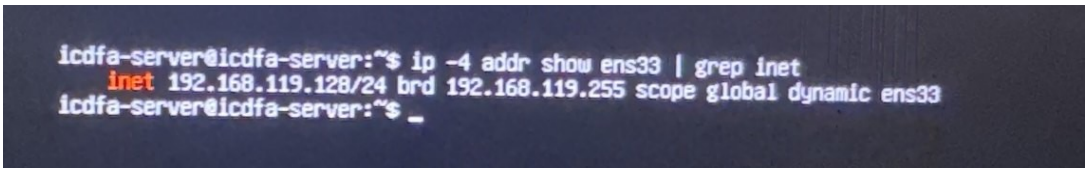
Ex. E33 | *not visible (18 Oct)* | *vm-console* | `apt-get install -y openssh-server` *(already the newest version) and* `systemctl enable --now ssh`. | *SHA-256 2e51a537b4191bcd…*



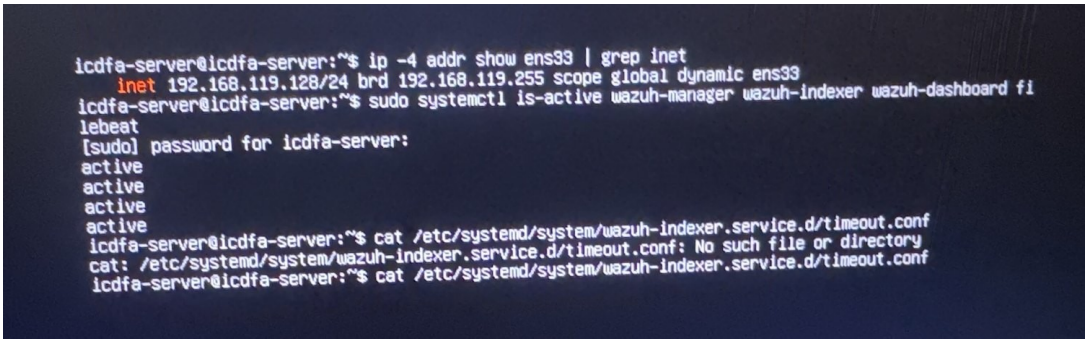
Ex. E34 | *not visible (18 Oct)* | *vm-console* | `ufw status` *inactive, sshd listening on port 22,* `ufw allow` *rules for 22, 443, 1514, 1515; status remains inactive.* | *SHA-256 6286dd7e1119532e…*



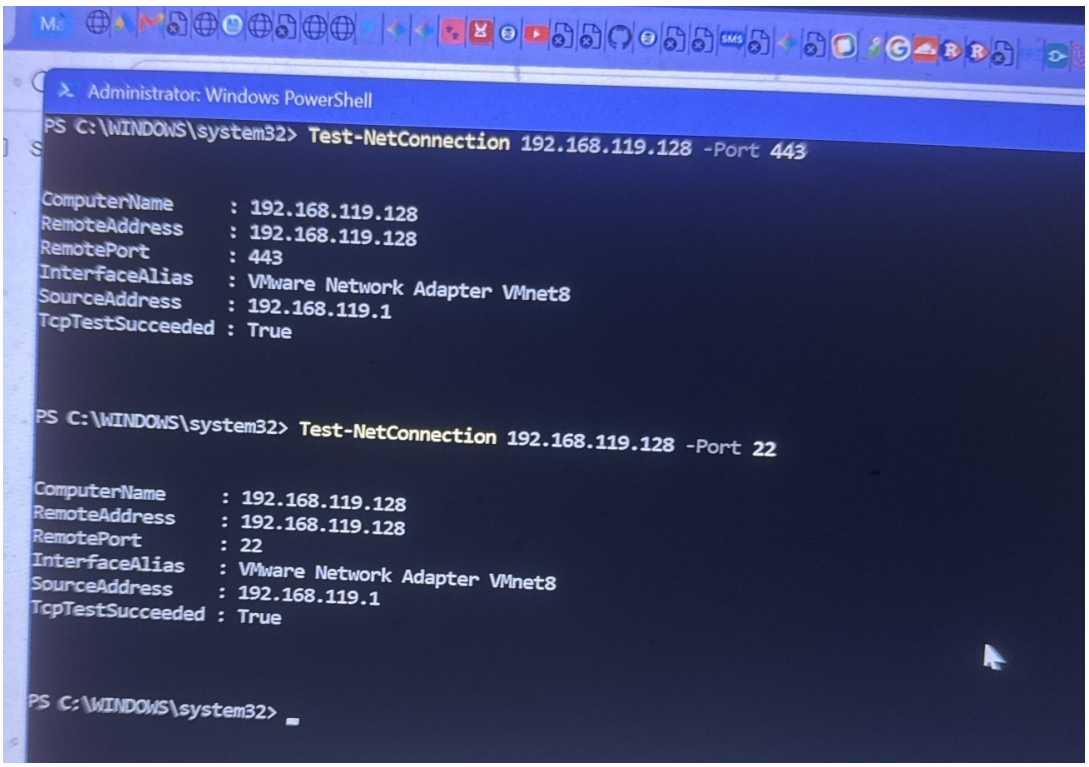
Ex. E35 | *not visible (18 Oct, after Restore Defaults)* | *powershell* | *Windows host: VMnet1 and VMnet8 are Up; VMnet8 192.168.119.1, VMnet1 192.168.225.1.* | *SHA-256 c96bc9e776538b0a…*



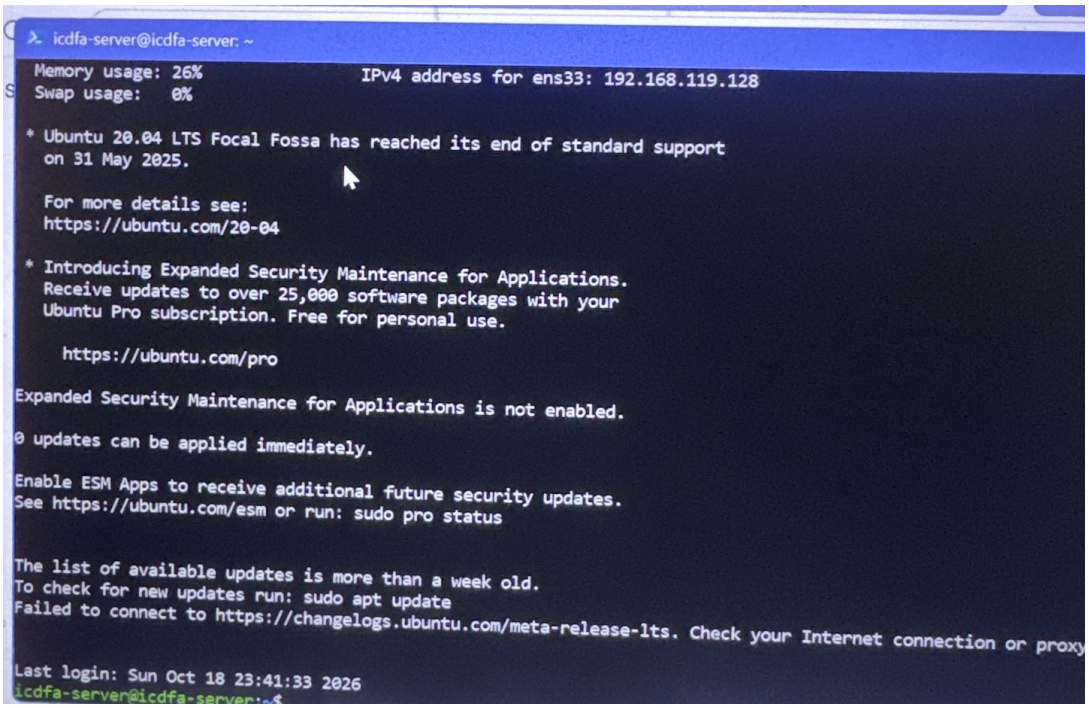
Ex. E36 | *not visible (18 Oct)* | *vm-console* | *VM: `ip -4 addr show ens33` gives 192.168.119.128/24 (dynamic).* | *SHA-256 41114246063b02c4…*



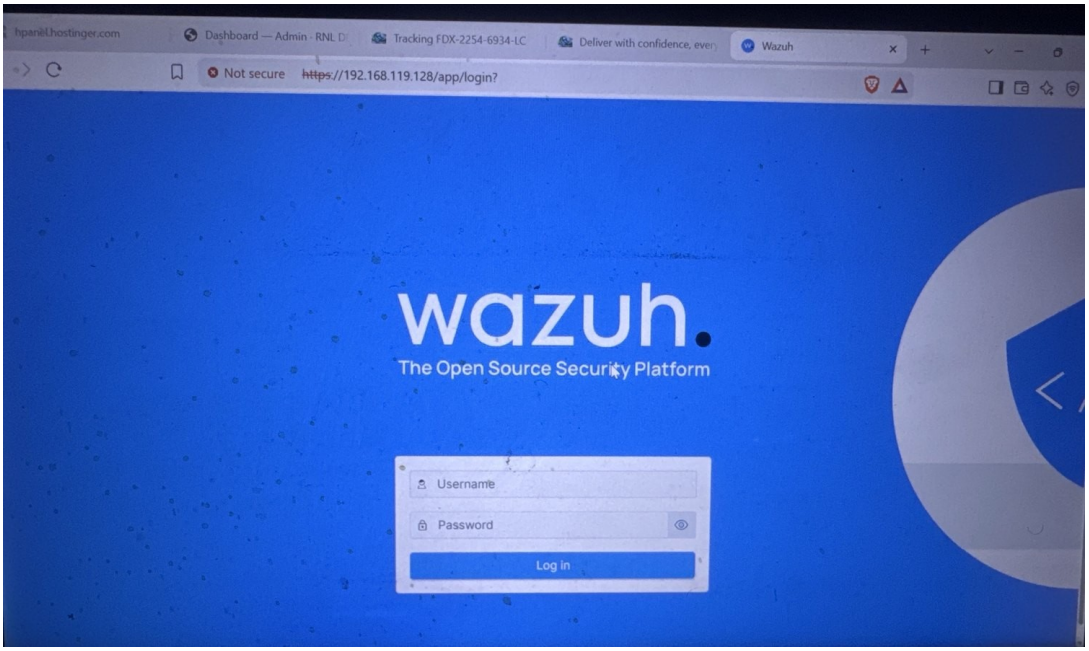
Ex. E37 | *not visible (18 Oct)* | *vm-console* | *VM: ens33 192.168.119.128; the four services are active; the indexer timeout drop-in file does not exist yet.* | *SHA-256 2d1b25fc6022ec6a…*



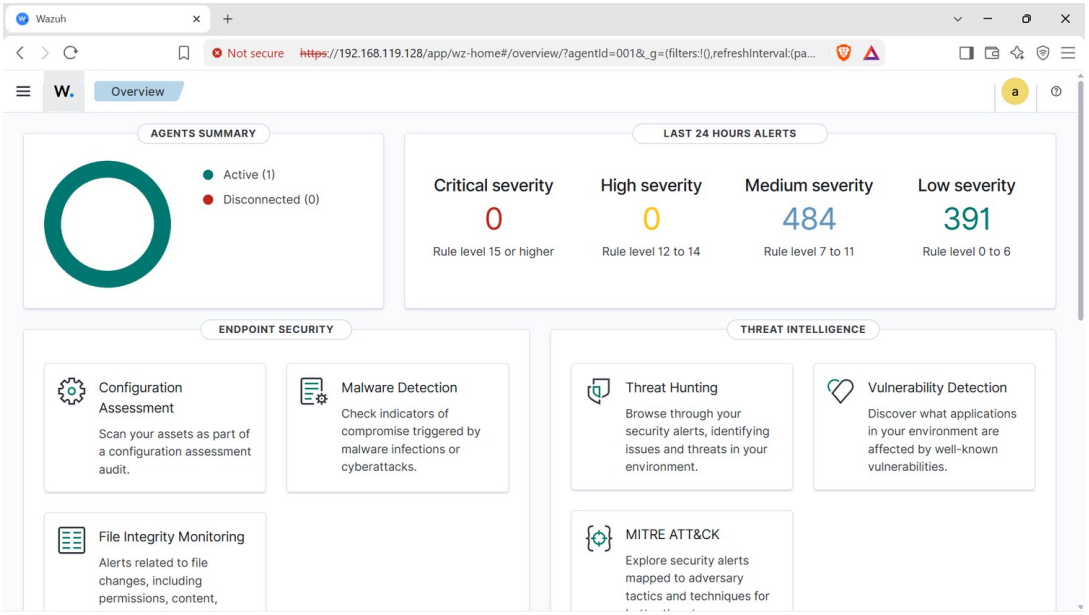
Ex. E38 | *not visible (18 Oct) | powershell | Windows host: `Test-NetConnection 192.168.119.128` succeeds on port 443 and on port 22 through VMware Network Adapter VMnet8 (source 192.168.119.1). | SHA-256 e2e8d74408877b33…*



Ex. E39 | *23:41:33 UTC last login (18 Oct) | powershell | SSH session to the VM from Windows: login banner shows "IPv4 address for ens33: 192.168.119.128"; memory 26%, swap 0%. | SHA-256 7c84e74235513a77…*



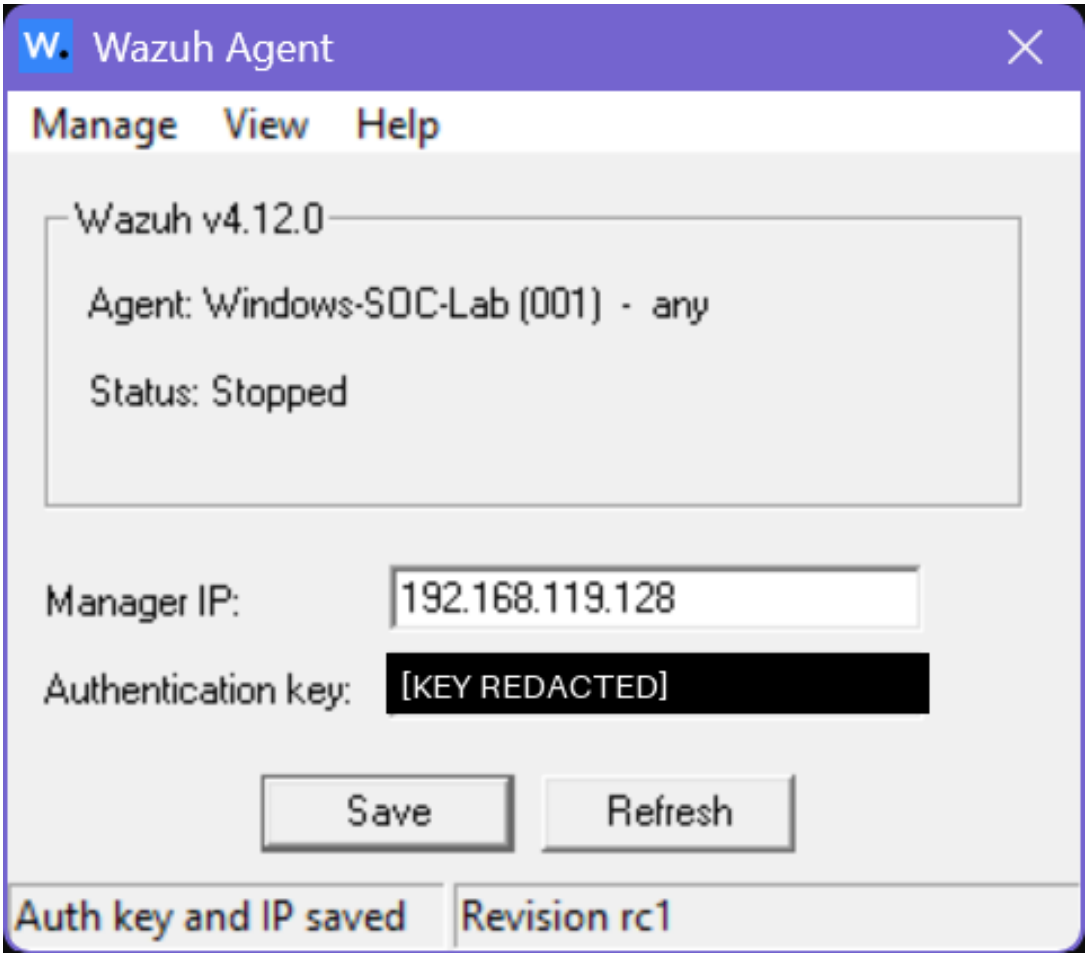
Ex. E40 | *not visible (18 Oct) | browser | Wazuh login page at https://192.168.119.128/app/login ("Not secure": self-signed certificate), shown before logging in. | SHA-256 d5fda30ff51ad31b…*



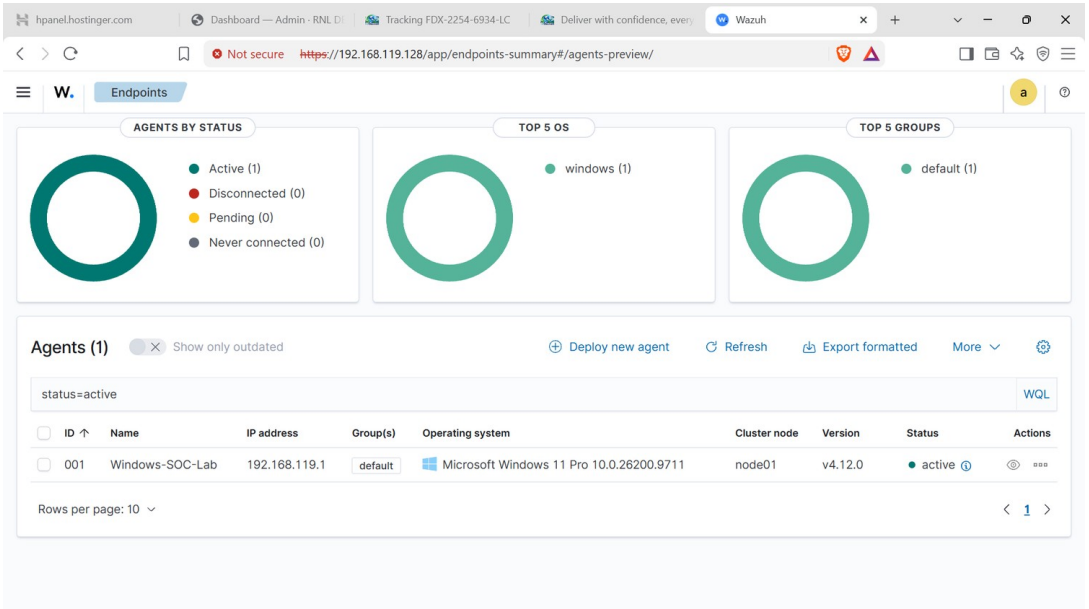
Ex. E41 | *about 09:13 local (16:13 UTC, 19 Oct) | dashboard | Wazuh Overview: agents summary Active (1) and Disconnected (0); last-24-hours alerts: critical 0, high 0, medium 484, low 391; File Integrity Monitoring module tile. | SHA-256 61976d1eb315f478…*

C. Agent registration

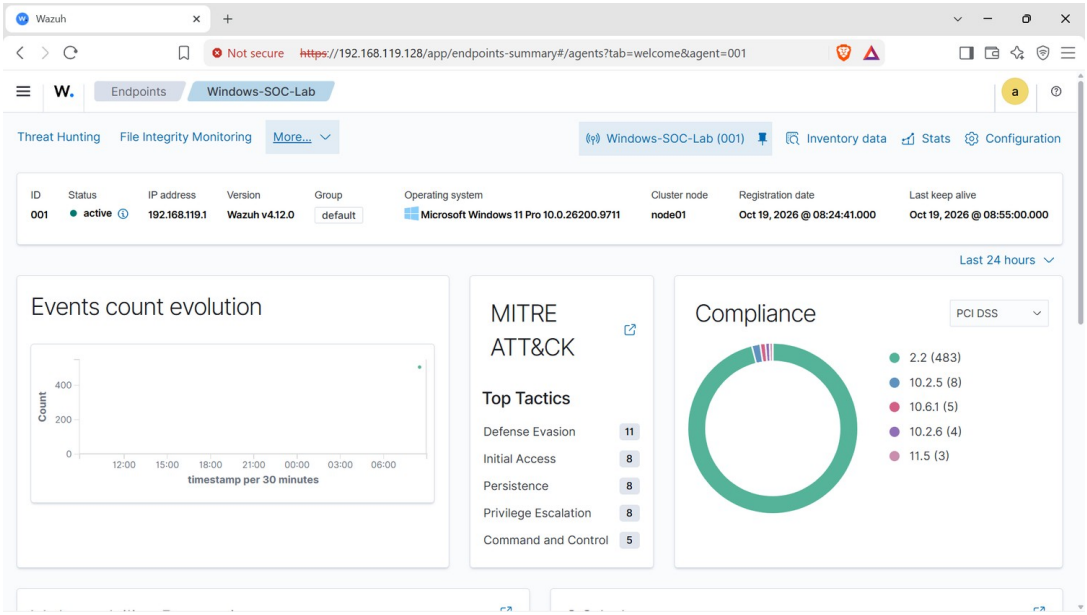
Agent Manager, dashboard agent list and agent overview.



Ex. E42 | *about 08:33 local (15:33 UTC, 19 Oct) | agent-gui | Wazuh Agent Manager on Windows: Agent Windows-SOC-Lab (001), Manager IP 192.168.119.128, "Auth key and IP saved", Status: Stopped. The key field is covered on purpose. | SHA-256 99dd4779ddbe7425…*



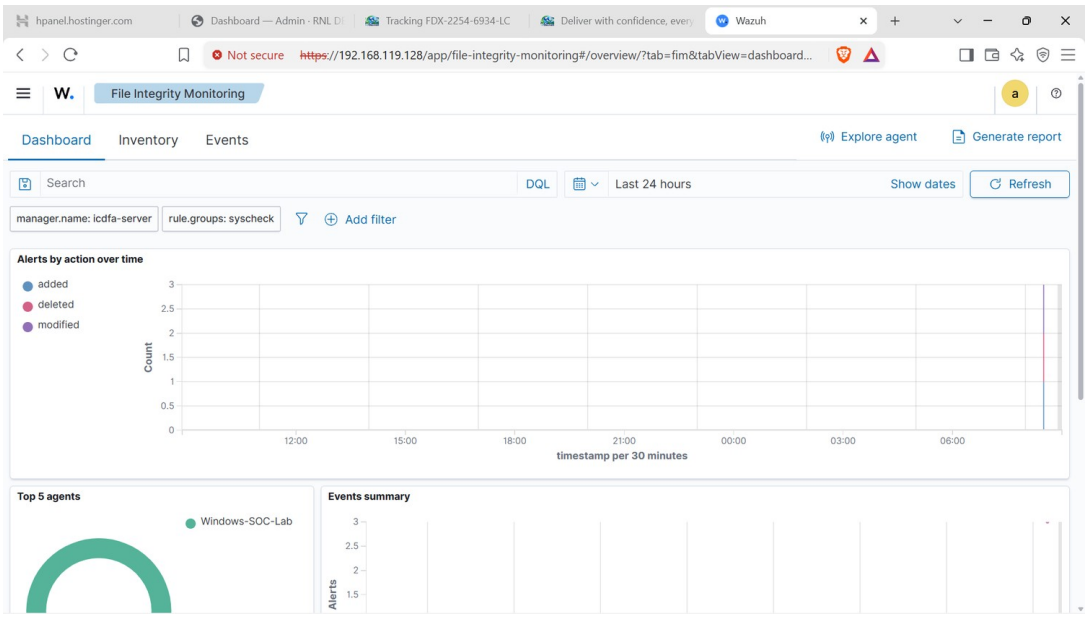
Ex. E43 | *about 08:37 local (15:37 UTC, 19 Oct) | dashboard | Dashboard Endpoints: Active (1); agent 001 Windows-SOC-Lab, IP 192.168.119.1, Windows 11 Pro 10.0.26200.9711, node01, v4.12.0, status active. | SHA-256 5660e90a0c0aa4eb…*



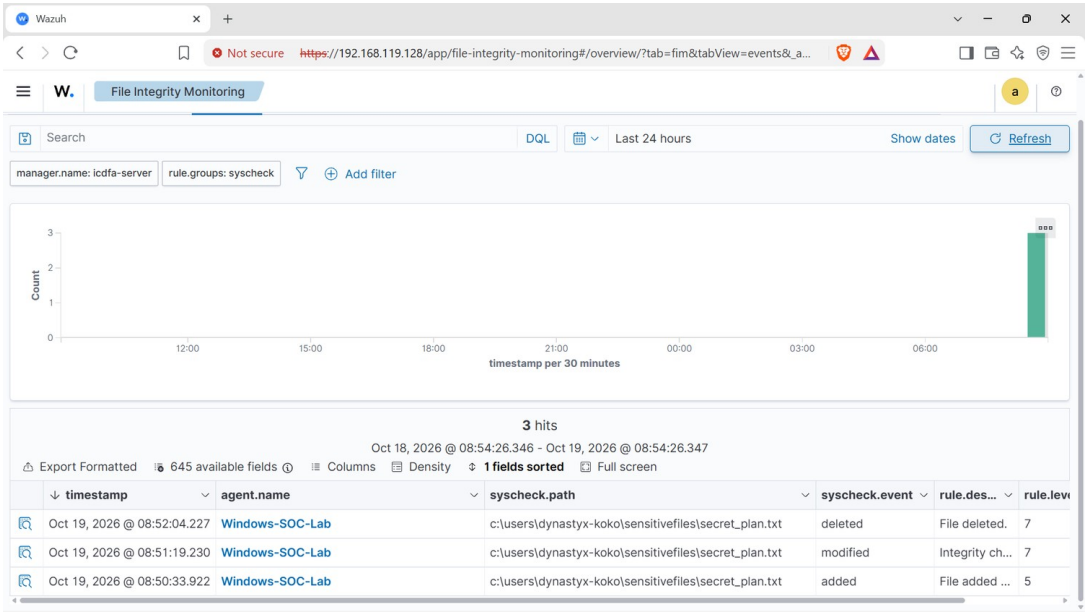
Ex. E44 | *08:55:00 local keep-alive (15:55 UTC, 19 Oct) | dashboard | Agent overview: registration date Oct 19 2026 08:24:41, last keep alive 08:55:00, status active. | SHA-256 93d54bc6f52d82e7…*

D. File integrity monitoring alerts

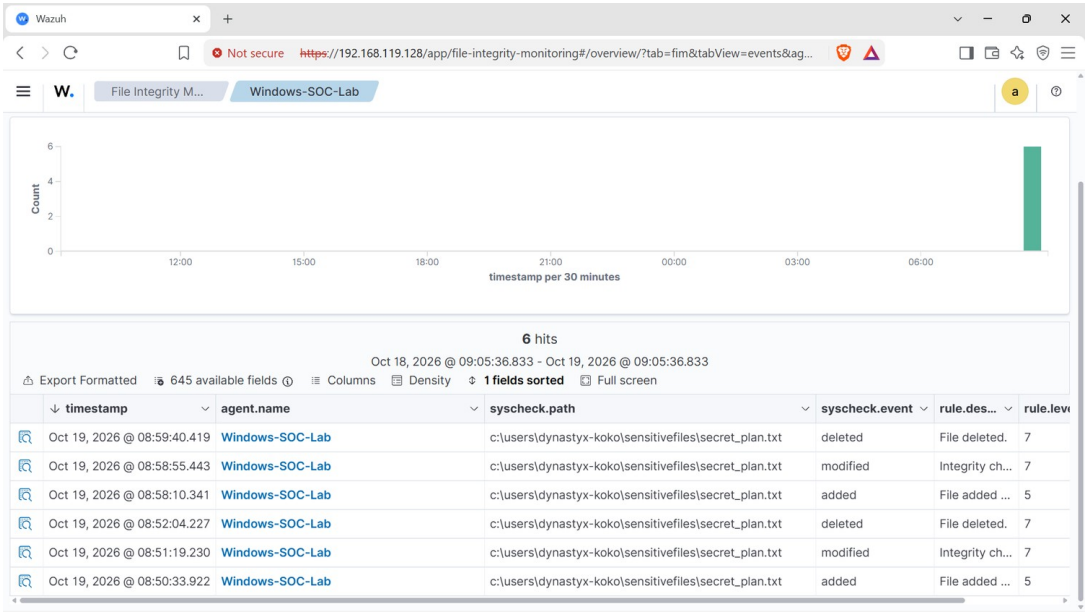
FIM dashboard and event lists for runs 1 and 2.



Ex. E45 | *08:54 local (15:54 UTC, 19 Oct) | dashboard | File Integrity Monitoring dashboard (last 24 hours): alerts by action (added, deleted, modified), top agent Windows-SOC-Lab; filters manager.name icdfa-server and rule.groups syscheck. | SHA-256 f9c7b5b99a79bb89…*



Ex. E46 | *08:50:33 to 08:52:04 local (15:50 to 15:52 UTC, 19 Oct) | dashboard | FIM Events, run 1: 3 hits for secret_plan.txt: 08:50:33.922 added (level 5), 08:51:19.230 modified (level 7), 08:52:04.227 deleted (level 7). | SHA-256 e3670aa20661a38f…*



Ex. E47 | *08:58:10 to 08:59:40 local (15:58 to 15:59 UTC, 19 Oct) | dashboard | FIM Events, runs 1 and 2: 6 hits; run 2 at 08:58:10.341 added, 08:58:55.443 modified, 08:59:40.419 deleted. | SHA-256 d9248fbd4d4ad501…*

T. Text exports

Files marked "rebuilt" were typed back from console text pasted by the analyst, because the VM and the Windows console offered no file export for them; the first two lines of each state this. Files marked "original upload" are unchanged copies. T11 is shown without the long win_perm_after block; the full file is in 04_Logs.

T01. T01_windows_network_diagnostics_before_fix.txt

Windows host diagnostics before the VMnet8 repair: ping and Test-NetConnection to 192.168.152.131 time out on Wi-Fi; VMnet8 shows "Not Present". (original upload 5c604fe7-attachment.txt (unchanged))

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> ping 192.168.152.131 -n 3
Pinging 192.168.152.131 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.152.131:
Packets: Sent = 3, Received = 0, Lost = 3 (100% loss),
PS C:\WINDOWS\system32> Test-NetConnection 192.168.152.131 -Port 443
WARNING: TCP connect to (192.168.152.131 : 443) failed
WARNING: Ping to 192.168.152.131 failed with status: TimedOut
ComputerName            : 192.168.152.131
RemoteAddress           : 192.168.152.131
RemotePort              : 443
InterfaceAlias          : Wi-Fi
SourceAddress           : 172.20.10.3
PingSucceeded           : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded        : False



PS C:\WINDOWS\system32> Test-NetConnection 192.168.152.131 -Port 22
WARNING: TCP connect to (192.168.152.131 : 22) failed
WARNING: Ping to 192.168.152.131 failed with status: TimedOut


ComputerName            : 192.168.152.131
RemoteAddress           : 192.168.152.131
RemotePort              : 22
InterfaceAlias          : Wi-Fi
SourceAddress           : 172.20.10.3
PingSucceeded           : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded        : False



PS C:\WINDOWS\system32> Get-NetAdapter | Where-Object { $_.Name -like "*VMnet*" -or $_.InterfaceDescription -like
"*VM

Name
Status      InterfaceDescription
----
------      --------------------
VMware Network Adapter VMnet1 Up          VMware Virtual Ethernet Adapter for VMnet1
VMware Network Adapter VMnet8 Not Present VMware Virtual Ethernet Adapter for VMnet8

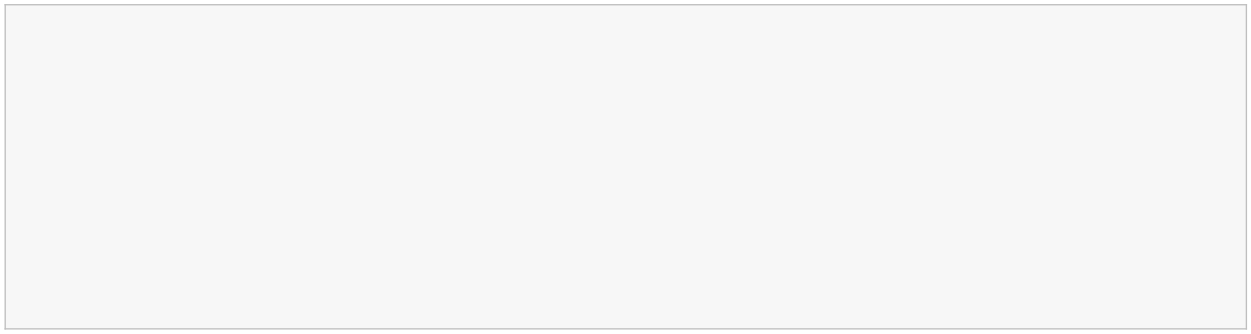

PS C:\WINDOWS\system32> Get-NetIPAddress -AddressFamily IPv4 | Where-Object { $_.IPAddress -like "192.168.152.*" } |
F
PS C:\WINDOWS\system32> Enable-NetAdapter -Name "VMware Network Adapter VMnet8" -Confirm:$false
PS C:\WINDOWS\system32> Get-NetConnectionProfile


Name
: iPhone xll
InterfaceAlias           : Wi-Fi
InterfaceIndex           : 15
NetworkCategory          : Public
```

```
DomainAuthenticationKind : None
IPv4Connectivity         : Internet
IPv6Connectivity         : NoTraffic



PS C:\WINDOWS\system32> Get-NetFirewallProfile | Format-Table Name, Enabled
```



T02. T02_manage_agents_transcript.txt

manage_agents session on the manager: agent 001 added, duplicate-name warning on a repeated attempt, key extracted (key removed), agent listed. (rebuilt from pasted console text; key replaced by a marker)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# Session on the Wazuh manager VM (icdfa-server), 2026/10/19 15:25 UTC. The agent key printed by option E is
replaced

icdfa-server@icdfa-server:~$ sudo /var/ossec/bin/manage_agents

****************************************
* Wazuh v4.12.0 Agent manager.         *
* The following options are available: *
****************************************
   (A)dd an agent (A).
   (E)xtract key for an agent (E).
   (L)ist already added agents (L).
   (R)emove an agent (R).
   (Q)uit.
Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
  Please provide the following:
   * A name for the new agent: Windows-SOC-Lab
   * The IP Address of the new agent: any
Confirm adding it?(y/n): y
Agent added with ID 001.

Choose your action: A,E,L,R or Q: A

- Adding a new agent (use '\q' to return to the main menu).
  Please provide the following:
   * A name for the new agent: Windows-SOC-Lab
   * The IP Address of the new agent: 001

** Invalid IP '001'. Please enter a valid IP Address.

   * The IP Address of the new agent: any
Confirm adding it?(y/n): y
2026/10/19 15:25:23 manage_agents: WARNING: 9008: Duplicate name

Choose your action: A,E,L,R or Q: e

Available agents:
   ID: 001, Name: Windows-SOC-Lab, IP: any
Provide the ID of the agent to extract the key (or '\q' to quit): 001

Agent key information for '001' is:
[AGENT KEY REMOVED FROM THIS EXPORT]

** Press ENTER to return to the main menu.

Choose your action: A,E,L,R or Q: q

manage_agents: Exiting.
icdfa-server@icdfa-server:~$ sudo /var/ossec/bin/manage_agents -l

Available agents:
   ID: 001, Name: Windows-SOC-Lab, IP: any

# Note: the second "A" was a repeated attempt; the prompt that appeared asked for an IP address, so "001" was
rejected
```

T03. T03_agent_control_list.txt

agent_control -l: agent 000 (server) and agent 001 Windows-SOC-Lab, Active. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
icdfa-server@icdfa-server:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: icdfa-server (server), IP: 127.0.0.1, Active/Local
   ID: 001, Name: Windows-SOC-Lab, IP: any, Active

List of agentless devices:
```

T04. T04_windows_agent_install_and_verify.txt

Agent installer download (5,406,720 bytes), silent install, service start, port 1514 test and "Connected to the server" log lines. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# Windows host, Administrator PowerShell, 2026-10-19 (host clock UTC-7)

PS C:\WINDOWS\system32> Start-BitsTransfer -Source "https://packages.wazuh.com/4.x/windows/wazuh-agent-4.12.0-1.msi"
-
PS C:\WINDOWS\system32> Get-Item "$env:USERPROFILE\Downloads\wazuh-agent-4.12.0-1.msi" | Select-Object Name, Length

Name                      Length
----                      ------
wazuh-agent-4.12.0-1.msi 5406720

PS C:\WINDOWS\system32> msiexec /i "$env:USERPROFILE\Downloads\wazuh-agent-4.12.0-1.msi" /q

# Wazuh Agent Manager (GUI): Manager IP 192.168.119.128 and the authentication key entered, then Save (see exhibit
E42

PS C:\WINDOWS\system32> Start-Service WazuhSvc
PS C:\WINDOWS\system32> Get-Service WazuhSvc

Status   Name               DisplayName
------   ----               -----------
Running  WazuhSvc           Wazuh

PS C:\WINDOWS\system32> Test-NetConnection 192.168.119.128 -Port 1514

ComputerName     : 192.168.119.128
RemoteAddress    : 192.168.119.128
RemotePort       : 1514
InterfaceAlias   : VMware Network Adapter VMnet8
SourceAddress    : 192.168.119.1
TcpTestSucceeded : True

PS C:\WINDOWS\system32> Start-Sleep -Seconds 20
PS C:\WINDOWS\system32> Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.log" -Pattern "Connected to
the

C:\Program Files (x86)\ossec-agent\ossec.log:77:2026/10/19 08:34:16 wazuh-agent: INFO: (4102): Connected to the
server
C:\Program Files (x86)\ossec-agent\ossec.log:167:2026/10/19 08:34:24 wazuh-agent: INFO: (4102): Connected to the
serve
```

T05. T05_fim_configuration_and_restart.txt

Folder creation, ossec.conf backup, the added directories line, service restart and the agent log lines confirming real-time monitoring. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# Windows host, Administrator PowerShell, 2026-10-19 (host clock UTC-7)

PS C:\WINDOWS\system32> New-Item -ItemType Directory -Force -Path "C:\Users\DynastyX-koko\SensitiveFiles"


    Directory: C:\Users\DynastyX-koko

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/19/2026   8:39 AM                SensitiveFiles

PS C:\WINDOWS\system32> $conf = "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS C:\WINDOWS\system32> Copy-Item $conf "$conf.bak"
PS C:\WINDOWS\system32> Get-ChildItem "C:\Program Files (x86)\ossec-agent\ossec.conf*"


    Directory: C:\Program Files (x86)\ossec-agent

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/19/2026   8:32 AM           9860 ossec.conf
-a----        10/19/2026   8:32 AM           9860 ossec.conf.bak

PS C:\WINDOWS\system32> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
# (line added inside <syscheck>, file saved)
PS C:\WINDOWS\system32> Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.conf" -Pattern
"SensitiveFiles"

C:\Program Files (x86)\ossec-agent\ossec.conf:156:    <directories realtime="yes" check_all="yes"
report_changes="yes"

PS C:\WINDOWS\system32> Restart-Service WazuhSvc
PS C:\WINDOWS\system32> Start-Sleep -Seconds 20
PS C:\WINDOWS\system32> Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.log" -Pattern "sensitivefiles|
Re

ossec.log:38:2026/10/19 08:34:16 wazuh-agent: INFO: (6003): Monitoring path: 'c:\programdata\microsoft\windows\start
m
ossec.log:130:2026/10/19 08:34:24 wazuh-agent: INFO: (6003): Monitoring path: 'c:\programdata\microsoft\windows\
start
ossec.log:180:2026/10/19 08:34:55 wazuh-agent: INFO: (6012): Real-time file integrity monitoring started.
ossec.log:224:2026/10/19 08:45:43 wazuh-agent: INFO: (6003): Monitoring path: 'c:\programdata\microsoft\windows\
start
ossec.log:225:2026/10/19 08:45:43 wazuh-agent: INFO: (6003): Monitoring path: 'c:\users\dynastyx-koko\
sensitivefiles',
```

T06. T06_compare_object_ossec_conf.txt

Compare-Object of ossec.conf.bak against ossec.conf: exactly one line differs. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
PS C:\WINDOWS\system32> $conf = "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS C:\WINDOWS\system32> Compare-Object (Get-Content "$conf.bak") (Get-Content $conf)

InputObject
-----------
    <directories realtime="yes" check_all="yes"
report_changes="yes">C:\Users\DynastyX-koko\SensitiveFiles</director..

# Exactly one line differs between the backup and the edited file. (The console cut the line at its right edge; the
fu
```

T07. T07_fim_run1_script_output.txt

Run 1 script output: the log file could not be written because the Documents folder does not exist; file operations themselves ran. (original upload a45e0d98-attachment.txt (unchanged))

```
PS C:\WINDOWS\system32> $dir = "C:\Users\DynastyX-koko\SensitiveFiles"
PS C:\WINDOWS\system32> $f   = "$dir\secret_plan.txt"
PS C:\WINDOWS\system32> $log = "$env:USERPROFILE\Documents\Lab05_FIM_events.txt"
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> function Mark($step) {
>>   $n = Get-Date
>>   "{0} | local {1} | UTC {2}" -f $step, $n.ToString("yyyy-MM-dd HH:mm:ss"), $n.ToUniversalTime().ToString("yyyy-
MM-
>> }
PS C:\WINDOWS\system32> function Show-Hash {
>>   "   SHA256={0}  size={1} bytes" -f (Get-FileHash $f -Algorithm SHA256).Hash, (Get-Item $f).Length | Tee-Object
-F
>> }
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Mark "1 CREATE"
out-file : Could not find file 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt'.
At line:3 char:3
+   "{0} | local {1} | UTC {2}" -f $step, $n.ToString("yyyy-MM-dd HH:mm ...
+   ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (:) [out-file], FileNotFoundException
    + FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\WINDOWS\system32> Set-Content -Path $f -Value "CONFIDENTIAL: Project plan v1"
PS C:\WINDOWS\system32> Show-Hash
out-file : Could not find file 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt'.
At line:2 char:3
+   "   SHA256={0}  size={1} bytes" -f (Get-FileHash $f -Algorithm SHA2 ...
+   ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (:) [out-file], FileNotFoundException
    + FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\WINDOWS\system32> Start-Sleep -Seconds 45
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Mark "2 MODIFY"
out-file : Could not find file 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt'.
At line:3 char:3
+   "{0} | local {1} | UTC {2}" -f $step, $n.ToString("yyyy-MM-dd HH:mm ...
+   ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (:) [out-file], FileNotFoundException
    + FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\WINDOWS\system32> Add-Content -Path $f -Value "Revision: budget increased, new target date added"
PS C:\WINDOWS\system32> Show-Hash
out-file : Could not find file 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt'.
At line:2 char:3
+   "   SHA256={0}  size={1} bytes" -f (Get-FileHash $f -Algorithm SHA2 ...
+   ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (:) [out-file], FileNotFoundException
    + FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\WINDOWS\system32> Start-Sleep -Seconds 45
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Mark "3 DELETE"
out-file : Could not find file 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt'.
At line:3 char:3
+   "{0} | local {1} | UTC {2}" -f $step, $n.ToString("yyyy-MM-dd HH:mm ...
+   ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (:) [out-file], FileNotFoundException
    + FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\WINDOWS\system32> Remove-Item -Path $f
PS C:\WINDOWS\system32> "   file still exists: $(Test-Path $f)" | Tee-Object -FilePath $log -Append
```

```
out-file : Could not find file 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt'.
At line:1 char:26
+ "   file still exists: $(Test-Path $f)" | Tee-Object -FilePath $log - ...
+                          ~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (:) [out-file], FileNotFoundException
    + FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\WINDOWS\system32> Get-Content "$env:USERPROFILE\Documents\Lab05_FIM_events.txt"
Get-Content : Cannot find path 'C:\Users\DynastyX-koko\Documents\Lab05_FIM_events.txt' because it does not exist.
At line:1 char:1
+ Get-Content "$env:USERPROFILE\Documents\Lab05_FIM_events.txt"
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (C:\Users\Dynast..._FIM_events.txt:String) [Get-Content],
ItemNotFoundEx
   ception
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetContentCommand

PS C:\WINDOWS\system32> Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.log" -Pattern "syscheck|
sensitiv

C:\Program Files (x86)\ossec-agent\ossec.log:225:2026/10/19 08:45:43 wazuh-agent: INFO: (6003): Monitoring path:
'c:\users\dynastyx-koko\sensitivefiles', with options 'size | permissions | owner | group | mtime | inode | hash_md5
|
hash_sha1 | hash_sha256 | attributes | report_changes | realtime'.


PS C:\WINDOWS\system32>
```

T08. T08_fim_run2_script_output.txt

Run 2 script output with timestamps (local and UTC), SHA-256 and size after create and modify. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# Run 2 (the logging-corrected run). Windows host, Administrator PowerShell, 2026-10-19 (host clock UTC-7).
# The function definitions are shown in the report (section 7). Output:

PS C:\WINDOWS\system32> Mark "1 CREATE"
1 CREATE | local 2026-10-19 08:58:10 | UTC 2026-10-19 15:58:10
PS C:\WINDOWS\system32> Set-Content -Path $f -Value "CONFIDENTIAL: Project plan v1"
PS C:\WINDOWS\system32> Show-Hash
   SHA256=482BC551AFE2DB96869B81591BA670607A158636D4211A8E0B8DCD34DE6C1A5D  size=31 bytes
PS C:\WINDOWS\system32> Start-Sleep -Seconds 45
PS C:\WINDOWS\system32> Mark "2 MODIFY"
2 MODIFY | local 2026-10-19 08:58:55 | UTC 2026-10-19 15:58:55
PS C:\WINDOWS\system32> Add-Content -Path $f -Value "Revision: budget increased, new target date added"
PS C:\WINDOWS\system32> Show-Hash
   SHA256=EADB58EBA2E7FF70A59FD9521CB523222B5F614BC7C6BD834A2BBFB99A13085C  size=82 bytes
PS C:\WINDOWS\system32> Start-Sleep -Seconds 45
PS C:\WINDOWS\system32> Mark "3 DELETE"
3 DELETE | local 2026-10-19 08:59:40 | UTC 2026-10-19 15:59:40
PS C:\WINDOWS\system32> Remove-Item -Path $f
PS C:\WINDOWS\system32> "   file still exists: $(Test-Path $f)" | Tee-Object -FilePath $log -Append
   file still exists: False
```

T09. T09_Lab05_FIM_run2_log.txt

Content of the run 2 record file Lab05_FIM_run2.txt. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# Content of C:\Users\DynastyX-koko\Lab05_Evidence\Lab05_FIM_run2.txt, printed with Get-Content after the run.

1 CREATE | local 2026-10-19 08:58:10 | UTC 2026-10-19 15:58:10
   SHA256=482BC551AFE2DB96869B81591BA670607A158636D4211A8E0B8DCD34DE6C1A5D  size=31 bytes
2 MODIFY | local 2026-10-19 08:58:55 | UTC 2026-10-19 15:58:55
   SHA256=EADB58EBA2E7FF70A59FD9521CB523222B5F614BC7C6BD834A2BBFB99A13085C  size=82 bytes
3 DELETE | local 2026-10-19 08:59:40 | UTC 2026-10-19 15:59:40
   file still exists: False
```

T10. T10_alert_554_added.json

Alert JSON, rule 554 (File added to the system), run 2, 15:58:10.341 UTC. (original upload 5202056d-attachment.txt (unchanged))

```
{
  "_index": "wazuh-alerts-4.x-2026.10.19",
  "_id": "UH_iVKEBBcEsBZMZhi6U",
  "_score": null,
  "_source": {
    "syscheck": {
      "mode": "realtime",
      "path": "c:\\users\\dynastyx-koko\\sensitivefiles\\secret_plan.txt",
      "sha1_after": "af9685f7be653653b608b4367a8a418c8fc81330",
      "uname_after": "Administrators",
      "mtime_after": "2026-10-19T15:58:10",
      "attrs_after": [
        "ARCHIVE"
      ],
      "size_after": "31",
      "uid_after": "S-1-5-32-544",
      "win_perm_after": [
        {
          "allowed": [
            "DELETE",
            "READ_CONTROL",
            "WRITE_DAC",
            "WRITE_OWNER",
            "SYNCHRONIZE",
            "READ_DATA",
            "WRITE_DATA",
            "APPEND_DATA",
            "READ_EA",
            "WRITE_EA",
            "EXECUTE",
            "READ_ATTRIBUTES",
            "WRITE_ATTRIBUTES",
            [
              "DELETE",
              "READ_CONTROL",
              "WRITE_DAC",
              "WRITE_OWNER",
              "SYNCHRONIZE",
              "READ_DATA",
              "WRITE_DATA",
              "APPEND_DATA",
              "READ_EA",
              "WRITE_EA",
              "EXECUTE",
              "READ_ATTRIBUTES",
              "WRITE_ATTRIBUTES"
            ],
            [
              "DELETE",
              "READ_CONTROL",
              "WRITE_DAC",
              "WRITE_OWNER",
              "SYNCHRONIZE",
              "READ_DATA",
              "WRITE_DATA",
              "APPEND_DATA",
              "READ_EA",
              "WRITE_EA",
              "EXECUTE",
              "READ_ATTRIBUTES",
              "WRITE_ATTRIBUTES"
            ]
```

```
        ],
        "name": "SYSTEM"
      },
      {
        "allowed": [
          "DELETE",
          "READ_CONTROL",
          "WRITE_DAC",
          "WRITE_OWNER",
          "SYNCHRONIZE",
          "READ_DATA",
          "WRITE_DATA",
          "APPEND_DATA",
          "READ_EA",
          "WRITE_EA",
          "EXECUTE",
          "READ_ATTRIBUTES",
          "WRITE_ATTRIBUTES"
        ],
        "name": "Administrators"
      },
      {
        "allowed": [
          "DELETE",
          "READ_CONTROL",
          "WRITE_DAC",
          "WRITE_OWNER",
          "SYNCHRONIZE",
          "READ_DATA",
          "WRITE_DATA",
          "APPEND_DATA",
          "READ_EA",
          "WRITE_EA",
          "EXECUTE",
          "READ_ATTRIBUTES",
          "WRITE_ATTRIBUTES"
        ],
        "name": "DynastyX-koko"
      }
    ],
    "event": "added",
    "md5_after": "c44b78358b834b1fa4760c96f5fd4ac9",
    "sha256_after": "482bc551afe2db96869b81591ba670607a158636d4211a8e0b8dcd34de6c1a5d"
  },
  "input": {
    "type": "log"
  },
  "agent": {
    "ip": "192.168.119.1",
    "name": "Windows-SOC-Lab",
    "id": "001"
  },
  "manager": {
    "name": "icdfa-server"
  },
  "rule": {
    "firedtimes": 2,
    "mail": false,
    "level": 5,
    "pci_dss": [
      "11.5"
    ],
```

```
      "hipaa": [
        "164.312.c.1",
        "164.312.c.2"
      ],
      "tsc": [
        "PI1.4",
        "PI1.5",
        "CC6.1",
        "CC6.8",
        "CC7.2",
        "CC7.3"
      ],
      "description": "File added to the system.",
      "groups": [
        "ossec",
        "syscheck",
        "syscheck_entry_added",
        "syscheck_file"
      ],
      "id": "554",
      "nist_800_53": [
        "SI.7"
      ],
      "gpg13": [
        "4.11"
      ],
      "gdpr": [
        "II_5.1.f"
      ]
    },
    "location": "syscheck",
    "decoder": {
      "name": "syscheck_new_entry"
    },
    "id": "1792425490.2257220",
    "full_log": "File 'c:\\users\\dynastyx-koko\\sensitivefiles\\secret_plan.txt' added\nMode: realtime\n",
    "timestamp": "2026-10-19T15:58:10.341+0000"
  },
  "fields": {
    "syscheck.mtime_after": [
      "2026-10-19T15:58:10.000Z"
    ],
    "timestamp": [
      "2026-10-19T15:58:10.341Z"
    ]
  },
  "sort": [
    1792425490341
  ]
}
```

T11. T11_alert_550_modified.json

Alert JSON, rule 550 (Integrity checksum changed), run 2, 15:58:55.443 UTC. (original upload 0065da04-attachment.txt (unchanged; 21456e6e is an identical copy))

```
{
  "_index": "wazuh-alerts-4.x-2026.10.19",
  "_id": "UX_jVKEBBcEsBZMZNi5F",
  "_score": null,
  "_source": {
    "syscheck": {
      "size_before": "31",
      "uname_after": "Administrators",
      "mtime_after": "2026-10-19T15:58:55",
      "size_after": "82",
      "md5_before": "c44b78358b834b1fa4760c96f5fd4ac9",
      "diff": "---\n> Revision: budget increased, new target date added\n",
      "win_perm_after": [
        {
          "allowed": [
            "DELETE",
            "READ_CONTROL",
            "WRITE_DAC",
            "WRITE_OWNER",
            "SYNCHRONIZE",
            "READ_DATA",
            "WRITE_DATA",
            "APPEND_DATA",
            "READ_EA",
            "WRITE_EA",
            "EXECUTE",
            "READ_ATTRIBUTES",
            "WRITE_ATTRIBUTES",
            [
              "DELETE",
              "READ_CONTROL",
              "WRITE_DAC",
              "WRITE_OWNER",
              "SYNCHRONIZE",
              "READ_DATA",
              "WRITE_DATA",
              "APPEND_DATA",
              "READ_EA",
              "WRITE_EA",
              "EXECUTE",
              "READ_ATTRIBUTES",
              "WRITE_ATTRIBUTES"
            ],
            [
              "DELETE",
              "READ_CONTROL",
              "WRITE_DAC",
              "WRITE_OWNER",
              "SYNCHRONIZE",
              "READ_DATA",
              "WRITE_DATA",
              "APPEND_DATA",
              "READ_EA",
              "WRITE_EA",
              "EXECUTE",
              "READ_ATTRIBUTES",
              "WRITE_ATTRIBUTES"
            ]
          ],
          "name": "SYSTEM"
        },
        {
```

```
        "allowed": [
          "DELETE",
          "READ_CONTROL",
          "WRITE_DAC",
          "WRITE_OWNER",
          "SYNCHRONIZE",
          "READ_DATA",
          "WRITE_DATA",
          "APPEND_DATA",
          "READ_EA",
          "WRITE_EA",
          "EXECUTE",
          "READ_ATTRIBUTES",
          "WRITE_ATTRIBUTES"
        ],
        "name": "Administrators"
      },
      {
        "allowed": [
          "DELETE",
          "READ_CONTROL",
          "WRITE_DAC",
          "WRITE_OWNER",
          "SYNCHRONIZE",
          "READ_DATA",
          "WRITE_DATA",
          "APPEND_DATA",
          "READ_EA",
          "WRITE_EA",
          "EXECUTE",
          "READ_ATTRIBUTES",
          "WRITE_ATTRIBUTES"
        ],
        "name": "DynastyX-koko"
      }
    ],
    "sha256_before": "482bc551afe2db96869b81591ba670607a158636d4211a8e0b8dcd34de6c1a5d",
    "mtime_before": "2026-10-19T15:58:10",
    "mode": "realtime",
    "path": "c:\\users\\dynastyx-koko\\sensitivefiles\\secret_plan.txt",
    "sha1_after": "0d455b03ec6c1a7aa3cade58cf747613ab79a51a",
    "changed_attributes": [
      "size",
      "mtime",
      "md5",
      "sha1",
      "sha256"
    ],
    "attrs_after": [
      "ARCHIVE"
    ],
    "uid_after": "S-1-5-32-544",
    "event": "modified",
    "md5_after": "6098bc161a45d5886d2815ab0e218861",
    "sha1_before": "af9685f7be653653b608b4367a8a418c8fc81330",
    "sha256_after": "eadb58eba2e7ff70a59fd9521cb523222b5f614bc7c6bd834a2bbfb99a13085c"
  },
  "input": {
    "type": "log"
  },
  "agent": {
    "ip": "192.168.119.1",
```

```
      "name": "Windows-SOC-Lab",
      "id": "001"
    },
    "manager": {
      "name": "icdfa-server"
    },
    "rule": {
      "mail": false,
      "level": 7,
      "pci_dss": [
        "11.5"
      ],
      "hipaa": [
        "164.312.c.1",
        "164.312.c.2"
      ],
      "tsc": [
        "PI1.4",
        "PI1.5",
        "CC6.1",
        "CC6.8",
        "CC7.2",
        "CC7.3"
      ],
      "description": "Integrity checksum changed.",
      "groups": [
        "ossec",
        "syscheck",
        "syscheck_entry_modified",
        "syscheck_file"
      ],
      "nist_800_53": [
        "SI.7"
      ],
      "gdpr": [
        "II_5.1.f"
      ],
      "firedtimes": 2,
      "mitre": {
        "technique": [
          "Stored Data Manipulation"
        ],
        "id": [
          "T1565.001"
        ],
        "tactic": [
          "Impact"
        ]
      },
      "id": "550",
      "gpg13": [
        "4.11"
      ]
    },
    "location": "syscheck",
    "decoder": {
      "name": "syscheck_integrity_changed"
    },
    "id": "1792425535.2258467",
    "full_log": "File 'c:\\users\\dynastyx-koko\\sensitivefiles\\secret_plan.txt' modified\nMode: realtime\nChanged
at
    "timestamp": "2026-10-19T15:58:55.443+0000"
```

```
  },
  "fields": {
    "syscheck.mtime_after": [
      "2026-10-19T15:58:55.000Z"
    ],
    "syscheck.mtime_before": [
      "2026-10-19T15:58:10.000Z"
    ],
    "timestamp": [
      "2026-10-19T15:58:55.443Z"
    ]
  },
  "sort": [
    1792425535443
  ]
}
```


T12. T12_manager_alerts_log_rule_ids.txt

Manager alerts.log search: two sequences of rules 554, 550, 553 with levels 5, 7, 7. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# On the Wazuh manager VM, 2026-10-19. The manager keeps its own plain-text copy of each alert in
/var/ossec/logs/aler
# (The first sudo password attempt was mistyped and repeated; the pasted text also contains remnants of a retyped
comm

icdfa-server@icdfa-server:~$ sudo grep -E "Rule: (550|553|554) " /var/ossec/logs/alerts/alerts.log | tail -n 8
Rule: 554 (level 5) -> 'File added to the system.'
Rule: 550 (level 7) -> 'Integrity checksum changed.'
Rule: 553 (level 7) -> 'File deleted.'
Rule: 554 (level 5) -> 'File added to the system.'
Rule: 550 (level 7) -> 'Integrity checksum changed.'
Rule: 553 (level 7) -> 'File deleted.'

# Two complete add, modify, delete sequences (run 1 and run 2), in the same order as the actions.
```

T13. T13_indexer_timeout_dropin.txt

systemd drop-in raising the indexer start timeout to 900 seconds. (rebuilt from pasted console text)

```
# Rebuilt from console text pasted by the analyst into the working chat (not a file copy). Only change: markers in
[sq
# Times: Windows host clock is UTC-7; Wazuh manager and VM use UTC.
# On the Wazuh manager VM, 2026-10-19 (after the earlier mistyped attempt, see exhibit E31)

icdfa-server@icdfa-server:~$ sudo mkdir -p /etc/systemd/system/wazuh-indexer.service.d
icdfa-server@icdfa-server:~$ printf '[Service]\nTimeoutStartSec=900\n' | sudo tee /etc/systemd/system/wazuh-
indexer.se
[Service]
TimeoutStartSec=900
icdfa-server@icdfa-server:~$ sudo systemctl daemon-reload
icdfa-server@icdfa-server:~$ cat /etc/systemd/system/wazuh-indexer.service.d/timeout.conf
[Service]
TimeoutStartSec=900
```